

Pivotal-Response Rederivation

Clean Two-Round Baseline Under Simultaneous Sealed Ballots

Formal model workspace

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1 Status and readback

This autonomous workspace rederives the clean two-round model by dependency-safe backward reasoning under **Perfect Bayesian equilibrium**. It does not add a global deletion of weakly dominated actions. A voter chooses yes at a payoff-relevant equality; when its action has zero probability of changing any payoff-relevant outcome, the assessment records an explicit completion without treating that completion as the substantive equilibrium object.

The conceptual method is:

1. reconstruct primitives and histories without importing candidate formulas;
2. define information sets, vote relevance, transitions, payoff dates, and assessment-level completion discipline;
3. validate the continuation DAG and obtain an independent read-only Gate 0 review;
4. solve and freeze the three independent terminal Round-2 problems in native Round-2 units;
5. import those interfaces literally into Round 1, applying **beta** exactly once where a terminal continuation enters a Round-1 comparison;
6. derive collective entry, institutional comparison, and a nonmigrating claim-level survival matrix only after their dependencies pass.

The artifacts include this document, versioned continuation interfaces, dependency-complete review bundles, detailed node proofs, the JSON DAG, four machine-readable Gate 0 registries, claim-level and verification tables, a protected-artifact hash manifest, a proof ledger, status reports, and separate R verifiers. The protected manuscripts, prior derivation, quarantined workspace and its historical reports, reviews, scripts, and outputs remain read-only. Their initial SHA-256 values are recorded in the protected-artifact manifest.

Analytic status: the complete DAG is **PASS** through the nonmigrating survival inventory at **passed_order=67**; the dependency checker reports **Ready: none**. Every substantive node was implemented by an agent distinct from its read-only reviewers. Rejected candidates remain in the ledger as provenance and are not consumed. The integrated document and its rendered outputs are a post-freeze presentation layer: they may summarize or reproduce the frozen objects, but they cannot repair or reinterpret them.

2 Provenance and quarantine

The implementation began from:

Field	Recorded value
Git root	repository PowerBayesianPersuasion
Branch	main
HEAD	df3a0f2...
Worktree at start	clean
Primitive source	power_architecture_derivations.Rmd
Quarantined history	undominated_voting_rederivation.Rmd

Field	Recorded value
Contract interface	<code>gate0_contract_v1.json</code>

The primitive source is used only to reconstruct the players, timing, contract space, budgets, information, payoffs, and continuation protocol. Its equilibrium formulas and classifications are not inputs. The quarantined file is provenance only and supplies no result, threshold, branch, or selection to this workspace.

Neither manuscript is edited or compiled here. No protected historical file is edited or compiled here.

3 Extensive-form contract

3.1 Players, type, prior, and primitive domain

There are $N = m + 1 \geq 3$ states: one hegemon H and m weak states $W_1, \dots, W_m$. Nature draws a persistent type

$$\theta \in \{0, 1\}$$

before formation. Only H observes the realization. Weak states share the common prior

$$\mu = \Pr(\theta = 1) \in [0, 1].$$

The weak institutional surplus is fixed, public, and normalized to one. Weak outside payoffs are zero. The clean primitive domain is

$$0 \leq o_0 < o_1 \leq \bar{y} \leq 1, \quad \beta \in (0, 1].$$

An agreement that includes H gives it y ; hence $b_0 = b_1 = 0$. The outside payoff o_θ is external to the unit institutional surplus. The rule $R \in \{U, M\}$ is fixed exogenously; there is no rule-choice move.

All primitives, the prior, the recognition rule, the contract space, the ballot, the transition map, payoffs, and the two tie rules below are common knowledge.

3.2 Formation, recognition, and horizon

After Nature draws the type, but before Round-1 recognition, the weak states make one collective all-or-nothing formation decision without observing θ . If they form, each pays an external sunk cost $\chi \geq 0$. If they do not form, every weak state receives zero and H receives o_θ .

Let α index a complete equilibrium assessment or an authorized selection from the downstream payoff correspondence. The gross collective value under rule R is

$$V_W^R(\alpha, \mu) = \frac{1}{m} \mathbb{E}_\alpha \left[\sum_{j=1}^m u_j^R \mid \text{formation} \right].$$

The net value is $V_W^R(\alpha, \mu) - \chi$. Formation occurs if and only if this quantity is weakly positive; equality leads to formation. If bargaining payoffs form a correspondence, entry is evaluated assessment by assessment. Payoff bounds may support a selection-free claim, but they never choose an assessment or define the formation rule.

There are exactly two bargaining rounds. Round 2 is terminal. In every reached round, exactly one weak proposer is drawn uniformly. The draws are independent across rounds and independent of type and public history. The hegemon is never recognized in this baseline: $\pi_H = 0$ in both rounds.

3.3 Proposal space and budgets

When H is active and weak state i is recognized, it proposes

$$s_i = (y, (x_j)_{j \neq i}), \quad 0 \leq y \leq \bar{y}, \quad x_j \geq 0.$$

If the current agreement includes H , the proposer's residual is

$$x_i^H = 1 - y - \sum_{j \neq i} x_j \geq 0.$$

If a majority agreement passes without H , the participation component is not burned. It is reabsorbed by the proposer:

$$x_i^{WO} = 1 - \sum_{j \neq i} x_j = x_i^H + y.$$

A named x_j is paid whenever the current proposal passes, even if j voted no. The participation component y is paid only when the implemented agreement includes H . After H has opted out, a weak-only proposal fixes $y = 0$ and allocates the unit weak surplus.

The proposal space contains gifts to voters whose ballots may not be needed for passage. They are not removed from the action space or from deviation checks.

3.4 Simultaneous ballot and quotas

The proposer is counted as voting yes. All other active players observe the public history and current proposal, then vote yes or no simultaneously. H also observes its private type. No voter observes another current vote when acting. The complete vector and outcome become public only after every vote has been cast.

In the absence of an authorized correlating device, behavioral randomizations are independent conditional on the player's information set. A derivation may not insert correlated ballots.

The quotas are

$$Q_U = N, \quad Q_M = q = \left\lfloor \frac{N}{2} \right\rfloor + 1.$$

After H opts out, the original quota is retained rather than recalculated. Because only $m = N - 1$ players then remain, the unanimity quota is impossible. That branch is a terminal outcome, not an artificial continuation game.

3.5 Immediate opt-out and continuation

A no vote by type θ of H is an immediate and irreversible opt-out. No current or future agreement can include it after that action. In Round 1 the action pays o_θ immediately, without discount. In Round 2 it pays o_θ at the Round-2 date.

If H votes yes in Round 1 but weak votes make the current ballot fail, H remains active and the appropriate active- H Round-2 state is reached. If H votes no under majority, a current weak-only agreement passes when the weak yes votes reach q ; otherwise the game reaches the terminal weak-only Round-2 majority problem. A Round-1 no vote by H under unanimity ends at the quota-impossible terminal outcome.

If an active- H Round-2 ballot fails, H receives o_θ in Round-2 units and weak states receive zero. If H opted out in Round 1, it receives no additional Round-2 flow.

3.6 Payoff dates and discounting

Round 2 is solved entirely in its native current units, with no internal discount factor. In the backward step, **beta is applied exactly once only when a frozen Round-2 value enters Round 1**. Immediate Round-1 opt-out is not discounted. A predecessor must import a literal frozen continuation interface; it may not rederive or repair that continuation locally.

3.7 Solution concept and beliefs

The solution concept is Perfect Bayesian equilibrium. An assessment specifies formation, proposal strategies, behavioral ballot strategies at every public proposal history, H 's type-contingent ballots, and beliefs wherever the type remains payoff-relevant. Strategies must be sequentially rational and beliefs must follow Bayes' rule wherever possible.

The full public history and the posterior are separate state objects. Continuation strategies may depend on the proposal and the complete revealed vote vector even when two histories induce the same posterior. At a zero-probability history, the required belief must be stated as part of the assessment; it may not be introduced inside a proof to suppress a branch.

The maintained weak-vote-passive belief discipline reflects that weak voters do not observe the type: a unilateral weak vote is not direct evidence about θ . A type-contingent vote by H may update beliefs once the ballot closes. After completed opt-out, the type is payoff-irrelevant to the weak-only continuation.

3.8 Direct response convention and proposal ties

When a player's vote changes a payoff-relevant event with positive probability, it votes yes when the expected payoff from yes is greater than or equal to the expected payoff from no. It votes no when the expected payoff from no is strictly larger. At equality with positive probability of relevance, the player votes yes.

When relevance has probability zero, the two actions are locally outcome-equivalent holding the other strategies fixed. The assessment must still record a complete behavioral profile, but that bookkeeping action is not a substantive equilibrium class. Section 4.4 explains why local equivalence does not license simultaneous replacement of several actions.

Only after maximizing the weak proposer's payoff, a proposal-payoff tie is resolved by minimizing H 's expected payoff. If both the proposer and H tie while downstream objects differ, the outcome remains a correspondence.

4 Vote relevance

4.1 Outcome signature

Fix player i 's ballot information set I , its action $a_i \in \{Y, N\}$, the other simultaneous actions a_{-i} , and the type when it remains relevant. Define the outcome signature

$$\Phi_I(a_i, a_{-i}, \theta) = (A, J_H, O_H, P, S),$$

where:

- A records approval or rejection;
- J_H records inclusion or exclusion of H ;
- O_H records opt-out status;
- P records the implemented payment vector and the destination of y ;
- S records the induced terminal outcome or the continuation outcome/interface under the frozen assessment.

If an action reaches a continuation, S is the action-specific object $\kappa_{C_2}(h_2)$, including its induced belief and payoff distribution. It is not a raw successor label or a raw vote vector. Thus, in Round 1, two actions that reach the same named C_2 class can remain relevant when they produce different full histories and $\kappa_{C_2}(h_2^Y) \neq \kappa_{C_2}(h_2^N)$.

The other-action profile is relevant for voter i exactly when

Table 2: Gate 0 vote-relevance registry.

relevance_id	round	rule	h_status	player_class	structural_condition
RV01	R1	U	active	weak_nonproposer_j	w_minus_j=m-1
RV02	R1	U	active	weak_nonproposer_j	never
RV03	R1	M	active	weak_nonproposer_j	w_minus_j=q-2 when q-2>=1
RV04	R1	M	active	weak_nonproposer_j	w_minus_j=q-1
RV05	R2	U	active	weak_nonproposer_j	w_minus_j=m-1
RV06	R2	U	active	weak_nonproposer_j	never
RV07	R2	M	active	weak_nonproposer_j	w_minus_j=q-2 when q-2>=1
RV08	R2	M	active	weak_nonproposer_j	w_minus_j=q-1
RV09	R2	M	absent	weak_nonproposer_j	w_minus_j=q-1
RH01	R1	U	active	H_type_theta	analyze separately for every z
RH02	R1	M	active	H_type_theta	analyze separately for every z
RH03	R2	U	active	H_type_theta	analyze separately for every z
RH04	R2	M	active	H_type_theta	analyze separately for every z

$$\Phi_I(Y, a_{-i}, \theta) \neq \Phi_I(N, a_{-i}, \theta).$$

This definition is conditional on the information set. It does not condition on a simultaneous action the voter has not observed.

4.2 Expected voting incentive

Let $R_i(I)$ be the set of relevant pairs (θ, a_{-i}) . The expected difference is

$$\Delta_i(I) = \sum_{(\theta, a_{-i}) \in R_i(I)} \Pr(\theta, a_{-i} \mid I) [u_i(Y, a_{-i}; \theta, I) - u_i(N, a_{-i}; \theta, I)].$$

Every profile outside $R_i(I)$ contributes exactly zero. If $\Pr(R_i(I) \mid I) > 0$, the prescribed response is yes when $\Delta_i(I) \geq 0$ and no when $\Delta_i(I) < 0$. This is an expected comparison over events the vote actually changes, not a blanket quota test.

For weak voters, relevant profiles normally occur at quota boundaries and can change current payment versus continuation. For H , relevance must be derived separately. Even when weak votes already meet a majority quota, changing H 's ballot can change inclusion, payment y , opt-out, or the continuation. Terminal opt-out status is part of the outcome signature even when terminal monetary payoffs happen to coincide.

4.3 Formal relevance registry

The authoritative registry is `tables/pivotal_response_relevance_registry_v1.csv`. It separates weak and hegemon ballots by round, rule, participation status, and information set.

The complete 14-column registry remains authoritative in the CSV; the table shows only the columns needed to audit the partition in print.

For a weak nonproposer j , let $w_{-j} \in \{1, \dots, m-1\}$ count weak yes votes other than j , including the proposer's automatic yes. The current-approval structural toggle occurs exactly in the following cases:

- active- H unanimity: $H = Y$ and $w_{-j} = m-1$; no weak toggle can restore unanimity after $H = N$;
- active- H majority with $H = Y$: $w_{-j} = q-2$, an empty set when $N = 3$ because then $q-2 = 0 < w_{-j}$;
- active- H majority with $H = N$: $w_{-j} = q-1$;
- weak-only majority: $w_{-j} = q-1$.

These quota cases do not exhaust Round-1 relevance when both actions fail. The action-specific full histories must still be passed through the frozen continuation interface. The verifier enumerates the structural cases for $N = 3, \dots, 60$, including the empty $N = 3$ majority-with- $H = Y$ case. The hegemon is analyzed separately for every $z \in \{1, \dots, m\}$; no weak-voter quota shortcut is applied to its inclusion, opt-out, payment, or continuation comparison.

Table 3: Gate 0 information sets and observations.

information_set_id	stage	movers	unobserved	successors
I00	Nature	Nature	future actions	formation or nonformation
I01	Formation	weak coalition	theta	R1 recognition or nonformation terminal
I02	R1 proposal	recognized weak proposer i	theta and future ballots	R1 ballot I03 and I04
I03	R1 ballot	weak voter j	theta and every simultaneous vote	terminal outcome or an R2 state
I04	R1 ballot	H	every simultaneous weak vote	terminal outcome or an R2 state
I05	R2 active-H proposal	recognized weak proposer i	theta and future ballots	R2 ballot I06 and I07
I06	R2 active-H ballot	weak voter j	theta and every simultaneous vote	terminal outcome
I07	R2 active-H ballot	H	every simultaneous weak vote	terminal outcome
I08	R2 weak-only proposal	recognized weak proposer i	simultaneous weak ballots	R2 ballot I09
I09	R2 weak-only ballot	weak voter j	every simultaneous weak vote	terminal outcome

4.4 Product-safe completion discipline

Local irrelevance is not closed under changing several actions jointly. Consider terminal unanimity with $N = 4$. The proposer and H vote yes and the two weak nonproposers vote no. Holding the other no fixed, either weak voter can switch alone without producing passage, so each action is locally irrelevant. If both switch to yes, the proposal passes. Terminal weak-only majority with $N = 4, q = 3$ has the same structure.

Accordingly, this Gate 0 does not impose a universal player-by-player completion. Every assessment records an explicit full behavioral profile at each proposal history. Two full completions may be quotiented only if they induce identical approval, H inclusion and opt-out, transfers, successor state, relevant belief distribution, and payoff distribution at every affected proposal and history.

Any later invariance claim must be product-safe: it must compare the induced joint distributions, not merely verify that each vote is individually irrelevant at one profile. If this proof fails, the different completions remain distinct outcome classes or a correspondence. Proposal optimality must also remain unchanged before a result is called completion-invariant.

5 Coalitions, gifts, and proposal optimality

No coalition-size restriction is a primitive or an additional equilibrium concept. Offers required to obtain support must be derived from ballot incentives. **Gifts to outsiders remain feasible deviations** and must be included in the recognized proposer’s best-response problem.

Nor does local irrelevance generally imply zero outsider gifts or a minimum winning representation. In terminal weak-only majority with $N = 5, q = 3$, an on-path proposal may have the proposer plus all three other weak voters voting yes. Each nonproposer is locally irrelevant against the other two yes votes, and the proposal passes. At a different, zero-gift proposal history, all three nonproposers may vote no; each unilateral switch remains locally irrelevant because proposer plus one yes still falls short of q , and the proposal fails. Deleting a gift changes the proposal information set and need not preserve its completion or the proposer’s payoff.

This finding is not confined to a knife-edge. An open set of feasible positive gift vectors can accompany the all-yes completion, while a distinct deviated proposal receives the all-no completion. Therefore a general zero-gift or minimum-coalition conclusion is rejected in this contract. A compact minimum-winning representation may be used only for a proved subclass in which passage is secured or full completion invariance is product-safe and proposer optimality removes every redundant transfer. Otherwise the proof must retain unrestricted proposal-contingent completions, positive gifts, oversized support, and any resulting correspondence.

6 Histories and information sets

The authoritative information-set registry is `tables/pivotal_response_information_sets_v1.csv`. It records what every mover observes, what remains unobserved, when actions become public, the feasible actions, successors, and the relevant belief object.

The complete 11-column information registry remains authoritative in the CSV.

The key safeguards are:

Table 4: Gate 0 exhaustive ballot transitions in native payoff units.

transition_id	round	rule	h_status	h_vote	z_range	current_outcome	successor
PR01	R1	U	active	Y	m	pass_with_H	terminal_current_agreement
PR02	R1	U	active	Y	1..m-1	fail_weak_caused	r2_unanimity_active_h
PR03	R1	U	active	N	1..m	quota_impossible_after_optout	terminal_quota_impossible_u_after_h_optout
PR04	R1	M	active	Y	q-1..m	pass_with_H	terminal_current_agreement
PR05	R1	M	active	Y	1..q-2	fail_weak_caused	r2_majority_active_h
PR06	R1	M	active	N	q..m	pass_weak_only	terminal_current_weak_only_agreement
PR07	R1	M	active	N	1..q-1	fail_then_weak_only_R2	r2_majority_weak_only
PR08	R2	U	active	Y	m	pass_with_H	terminal_current_agreement
PR09	R2	U	active	Y	1..m-1	terminal_failure	terminal_failure_active_h
PR10	R2	U	active	N	1..m	quota_impossible_after_optout	terminal_quota_impossible_u_after_h_optout
PR11	R2	M	active	Y	q-1..m	pass_with_H	terminal_current_agreement
PR12	R2	M	active	Y	1..q-2	terminal_failure	terminal_failure_active_h
PR13	R2	M	active	N	q..m	pass_weak_only	terminal_current_weak_only_agreement
PR14	R2	M	active	N	1..q-1	terminal_failure_after_optout	terminal_failure_after_h_optout
PR15	R2	M	absent	NA	q..m	pass_weak_only	terminal_current_weak_only_agreement
PR16	R2	M	absent	NA	1..q-1	terminal_failure_weak_only	terminal_failure_weak_only

Table 5: Gate 0 payoff- and belief-sufficient states.

state_id	round	rule	h_status	stage	interface_role
entry_pre_recognition_R	pre-R1	R in {U;M}	active	formation and pre-recognition	Entry evaluates $V \cdot W^R(\alpha; \mu)$ assessment by assessment; bounds are inputs only to selection-free claims.
r1_pre_recognition_R	R1	R in {U;M}	active	pre-recognition	Root of the rule-specific R1 continuation before recognition.
r1_proposer_R	R1	R in {U;M}	active	proposer before proposal	Proposal strategy domain; s_i is an action and is not inside the pre-action state.
r1_ballot_weak_R	R1	R in {U;M}	active	weak ballot	Consumes the full action-specific C2 mapping $\kappa_{C2}(h_2)$ when current failure continues.
r1_ballot_h_R	R1	R in {U;M}	active	H ballot	Consumes action-specific $\kappa_{C2}(h_2)$ when H-yes continues; H-no is immediate optout.
r2_u_active_pre_recognition	R2	U	active	pre-recognition	This is C2_U_active(h2); the correspondence imported by R1 and it integrates recognition.
r2_u_active_proposer	R2	U	active	proposer before proposal	Internal node of C2_U_active(h2).
r2_u_active_ballot_weak	R2	U	active	weak ballot	Internal terminal ballot of C2_U_active(h2).
r2_u_active_ballot_h	R2	U	active	H ballot	Internal terminal ballot of C2_U_active(h2).
r2_m_active_pre_recognition	R2	M	active	pre-recognition	This is C2_M_active(h2); the correspondence imported by R1 and it integrates recognition.
r2_m_active_proposer	R2	M	active	proposer before proposal	Internal node of C2_M_active(h2).
r2_m_active_ballot_weak	R2	M	active	weak ballot	Internal terminal ballot of C2_M_active(h2).
r2_m_active_ballot_h	R2	M	active	H ballot	Internal terminal ballot of C2_M_active(h2).
r2_m_weak_only_pre_recognition	R2	M	absent	pre-recognition	This is C2_M_weak_only(h2); the correspondence imported by R1 and it integrates recognition.
r2_m_weak_only_proposer	R2	M	absent	proposer before proposal	Internal node of C2_M_weak_only(h2).
r2_m_weak_only_ballot_weak	R2	M	absent	weak ballot	Internal terminal ballot of C2_M_weak_only(h2).
terminal_outcome	terminal	R in {U;M}	active or absent	terminal	Terminal outcomes are not continuation-game nodes.

- weak ballot strategies cannot condition on the type or another current vote;
- H 's strategy can condition on its type but not on current weak votes;
- a Round-2 state retains the complete public Round-1 proposal and vote vector separately from its posterior;
- the weak-only Round-2 state requires no belief about the type for its payoff problem.

7 Exhaustive transition registry

Let $z \in \{1, \dots, m\}$ be the number of weak yes votes, including the proposer's automatic yes. The authoritative transition registry is `tables/pivotal_response_gate0_transitions_v1.csv`.

The complete 16-column transition registry, including native payoffs and discount notes, remains authoritative in the CSV.

The R verifier enumerates every admissible ballot tuple for $N = 3, \dots, 60$, both rules, both rounds, active- H yes and no, and the terminal weak-only majority state. Each tuple must map to exactly one of PR01–PR16. The symbolic range $1, \dots, q - 2$ is empty at $q = 2$, rather than new histories.

No terminal outcome is promoted to a decision problem. In particular, after an H -no under unanimity, the original quota is impossible and the branch ends.

8 Sufficient states

The authoritative sufficient-state registry is `tables/pivotal_response_sufficient_states_v1.csv`.

The complete 12-column state registry remains authoritative in the CSV.

The registry separates pre-recognition, proposer-before- s_i , weak-ballot, and H -ballot states. A weak-ballot key retains both voter j and the public proposal s_i ; an H -ballot key retains s_i and private θ . Each $C_2(h_2)$ imported

by Round 1 is a pre-recognition correspondence indexed by the full public h_2 and integrates the uniform proposer draw. Histories may be compressed only after proving that feasible actions, payoff consequences, beliefs, admissible strategies, and all future continuations coincide.

9 Continuation DAG and barriers

The machine-readable graph is `model_redesign/pivotal_response_game_dag.json`. Its dependency direction is

```
Gate 0
-> {R2-U-active, R2-M-active, R2-M-weak-only}
-> R2 batch review
-> {R1-U, R1-M}
-> R1 batch freeze
-> {entry-U, entry-M}
-> institutional comparison
-> nonmigrating survival matrix.
```

Every brace is an antichain conditional on the preceding reviewed freeze. The final graph has all nodes at `pass`; the generic checker validates the execution orders and reports `Ready: none`.

The dependency-first schedule is:

Batch	Nodes	Frozen closure	Status
Gate 0	<code>gate0_contract</code>	bundle 6e28cb3...	PASS, event 10
R2 antichain	three terminal nodes	batch 00f84d42...	PASS, event 25
R1 antichain	<code>r1_unanimity</code> ; <code>r1_majority</code>	batch f4ac7b89...	PASS, event 40
Entry	<code>entry_unanimity</code> ; <code>entry_majority</code>	batch 8817a9c5...	PASS, event 53
Comparison	comparison node	review 0acd9648...	PASS, event 60
Survival inventory	<code>v6_survival_matrix</code>	review 80c912ba...	PASS, event 67; no migration

Entry consumes its corresponding Round-1 payoff correspondence assessment by assessment; a valid bound supports only a selection-free claim. Institutional comparison uses the Cartesian product of complete entry assessments on the full common existence domain. No scalar selection, rank matching, or cross-rule randomization is inserted.

The following are terminal outcomes, not nodes: nonformation, current passage with H , current weak-only passage, terminal active- H failure, terminal weak-only failure, and the quota-impossible unanimity outcome after opt-out.

10 Continuation interfaces and native-time ledger

The versioning policy is in `model_redesign/pivotal_response_interfaces/README.md`. Every closed continuation states strategies, beliefs, payoff vectors or correspondences, native payoff date, existence and multiplicity, boundaries, equality cases, assumptions, checks, status, provenance, and dependency hashes. Candidate interfaces are immutable snapshots; authority to consume them is supplied by their reviewed batch bundle.

Imported object	Native date	Consuming date	Permitted transformation	Status
$C_{2,U}^A(h_2)$	R2	R1	$\beta C_{2,U}^A(h_2)$ exactly once	frozen f3ca4ebf...
$C_{2,M}^A(h_2)$	R2	R1	$\beta C_{2,M}^A(h_2)$ exactly once	frozen a4ae73e9...
$C_{2,M}^{WO}(h_2)$	R2	R1 weak payoffs only	$\beta C_{2,M}^{WO}(h_2)$ exactly once	frozen e0ec6cd3...

Imported object	Native date	Consuming date	Permitted transformation	Status
immediate o_θ after H -no in R1	R1	R1	none	primitive

Changing any frozen component changes its bundle hash, reopens that node, and invalidates every descendant derivation and review. A predecessor may not fill an interface gap locally.

The principal frozen objects and local check snapshots are:

Object	SHA-256 prefix	Checks	Independent closure
Gate 0	6e28cb3f...	69/69	two rereviews PASS
R2-U, active H	f3ca4ebf...	18/18	two reviews PASS
R2-M, active H	a4ae73e9...	17/17	two rereviews PASS
R2-M, weak-only	e0ec6cd3...	13/13	two reviews PASS
R2 batch	00f84d42...	30/30	frozen PASS
R1-U	37aae1bf...	37/37	two rereviews PASS
R1-M	21c3a9dd...	23/23	two rereviews PASS
R1 batch	f4ac7b89...	24/24	frozen PASS
Entry-U	05e39ad2...	37/37	two rereviews PASS
Entry-M	4eb343ce...	24/24	two reviews PASS
Entry batch	8817a9c5...	27/27	frozen PASS
Comparison	cab69c5d...	42/42	two reviews PASS
Comparison review	0acd9648...	26/26	frozen PASS
Survival candidate	5278b14d...	43/43	two reviews PASS
Survival review	80c912ba...	24/24	frozen PASS

Full digests are machine-readable in the corresponding review bundles and in the proof ledger. Prefixes are shown here only to keep the rendered table legible.

11 Proof ledger

Table 9: Pivotal-response proof and execution ledger (display columns).

object	status	started_order	passed_order
Initial Git and protected-artifact snapshot	checked numerically	1	2
Extensive-form contract and dependency-complete bundle	proved	5	10
Vote-relevance and completion discipline	proved	5	10
General zero-gift or minimal-coalition conclusion	rejected	1	NA
Exhaustive ballot transitions	proved	5	10
Sufficient states and information sets	proved	5	10
Continuation DAG and batch barriers	proved	5	10
Gate 0 review round 1	rejected	3	NA
Independent Gate 0 rereview and close	proved	7	10
Terminal unanimity with H active	proved	11	18
Terminal majority with H active, initial candidate	rejected	12	NA
Terminal majority with H active, repaired candidate	proved	19	22
Terminal majority weak-only	proved	13	18
R2 batch independent review and freeze	proved	23	25
Round 1 unanimity, initial candidate	rejected	26	NA
Round 1 unanimity, repaired candidate	proved	32	37
Round 1 majority, initial candidate	rejected	27	NA
Round 1 majority, repaired candidate	proved	33	37
R1 batch freeze	proved	38	40
Entry under unanimity, initial candidate	rejected	41	NA
Entry under unanimity, repaired candidate	proved	47	50
Entry under majority	proved	42	46
Entry batch independent review and freeze	proved	51	53
Institutional comparison	proved	54	60
v6 survival-matrix candidate	checked numerically	61	62

The ledger distinguishes analytic proof, numerical or mechanical checking, rejected candidates, and final reviewed freezes. Numerical enumeration does not substitute for the propositions below. The general zero-gift or minimum-coalition conclusion remains **rejected** by the proposal-contingent completion counterexample. Initial R2-M-active, R1-U, R1-M, and entry-U candidates are also retained as rejected provenance; only their repaired, independently approved hashes are consumed downstream.

12 Independent Gate 0 review closure

The formal-model and adversarial read-only reviewers independently answered all of the following questions affirmatively for the exact frozen bundle:

1. PR01–PR16 exhaust and partition the feasible ballot profiles;
2. the information sets prevent conditioning on unobserved current votes;
3. vote relevance includes approval, participation, opt-out, payments, and the induced terminal or assessed continuation outcome without using a raw successor label or collapsing H 's problem to quota pivotality;
4. equality is resolved as yes only when relevance has positive probability;
5. assessment-level completions and independent behavioral mixing are defined without overclaiming invariance from individual local irrelevance;
6. the $N=4$ joint-completion and $N=5$ gift counterexamples are handled correctly;
7. the unrestricted proposal space preserves gifts as deviations and avoids a general minimum-coalition claim;
8. the DAG enforces the Gate 0, R2-batch-to-R1, and R1-batch-to-entry barriers;
9. all Round-2 payoffs are native-date objects and discounting occurs exactly once upon import to Round 1;
10. no equilibrium result has entered the Gate 0 contract.

The first review round returned REPAIR on contract hash `37bd422d...`. After repair, both rereviews returned PASS with zero critical, major, or minor findings on bundle `sha256:6e28cb3f...`. The closure and conditions are recorded in `gate0_review_v1.json` and the independent Gate 0 review report. No unresolved Gate 0 protocol choice remains. Any bundle-component change reopens the gate and invalidates all descendants.

13 How to read the frozen node proofs

The remaining chapters reproduce the detailed node proofs that generated the reviewed interfaces. Proposition numbering is local to each named node. The notes are immutable implementation snapshots, so a note may describe its own interface as a review candidate; the batch and review bundles listed above are the authoritative records that subsequently accepted those exact bytes. The embedded notes do not select an equilibrium from a correspondence and do not authorize manuscript migration.

14 Round 2: terminal continuation correspondences

Each terminal problem is solved in Round-2 units. There is no **beta** inside a terminal interface.

14.1 Round 2 under unanimity with an active hegemon

14.1.1 Status and provenance

- Node: `r2_unanimity_active_h`.
- Solution concept: Perfect Bayesian equilibrium (PBE).
- Frozen dependency: `gate0_bundle_v1.json`, `sha256:6e28cb3f...`
- Native payoff date: Round 2.
- Discounting inside this node: none.
- Post-freeze status: exact interface accepted by two independent read-only reviews.

This derivation uses only the frozen Gate 0 primitives and registries. It does not import an equilibrium formula or classification from a historical derivation. In particular, it does not delete weakly dominated actions, impose a coalition size, remove gifts, or identify several locally irrelevant votes with one joint completion.

14.1.2 The terminal decision problem

There are $N=m+1 \geq 3$ players: the active hegemon H and m weak states. Weak state i is the recognized proposer. It is counted as voting yes. The remaining

$$n = m-1 = N-2 \geq 1$$

weak states and H vote simultaneously. A proposal is

$$\begin{aligned} s_i &= (y, (x_j)_{j \neq i}), \\ 0 &\leq y \leq \bar{y}, \\ x_j &\geq 0, \\ r_i &= 1 - y - \sum_{j \neq i} x_j \geq 0. \end{aligned}$$

Unanimity passes only if H and every weak nonproposer vote yes. Passage pays y to H , x_j to weak nonproposer j , and r_i to the proposer. Every failure pays zero to all weak states and o_θ to type θ of H . An H -no additionally implements immediate opt-out; an H -yes followed by a weak-caused failure does not. This opt-out distinction is part of the Gate 0 outcome signature even though the two failure branches give H the same monetary payoff.

Let ν be the public belief before proposal recognition, and let $\rho(s)$ be the weak voters' belief at the ballot following proposal s . Because the weak proposer has no private information, every proposal in the support of its strategy has $\rho(s) = \nu$. At a globally zero-probability proposal, PBE permits the assessment to state any $\rho(s)$ in $[0, 1]$. A deviating proposer nevertheless evaluates its deviation using the true preproposal distribution ν .

14.1.3 Exact ballot correspondence at a fixed proposal

For weak nonproposer j , let p_j in $[0, 1]$ be its independent behavioral probability of yes. Define

$$\begin{aligned} P_W &= \prod_j p_j, \\ h_\theta &\text{ in } \{0, 1\} = H \text{ type } \theta \text{'s yes action,} \\ a_\rho &= (1 - \rho)h_0 + \rho h_1. \end{aligned}$$

No correlation device is used.

14.1.3.1 Lemma 1 (hegemon response) At every fixed proposal and weak behavioral profile,

$$h_\theta = 1\{P_W = 0 \text{ or } y \geq o_\theta\}.$$

Proof. If all weak nonproposers vote yes, an H -yes changes its payoff from o_θ to y . Otherwise an H -yes ends in weak-caused failure and pays o_θ , just like H -no. Hence the monetary difference is

$$\Delta_H(\theta) = P_W (y - o_\theta).$$

Changing H 's action changes opt-out status at every weak vote profile, so relevance has positive probability even when $P_W = 0$. The Gate 0 equality rule then selects yes at a zero monetary difference. If $P_W > 0$, it selects yes exactly when $y \geq o_\theta$. There is no type mixing at either threshold.

14.1.3.2 Lemma 2 (weak response) For weak nonproposer j , let $P_{-j} = \prod_{k \neq j} p_k$. Its vote is relevant with probability $a_\rho P_{-j}$. On this event, yes pays $x_j \geq 0$ and no pays zero. Thus

$$\begin{aligned} &\text{if } a_\rho P_{-j} > 0, \text{ then } p_j = 1; \\ &\text{if } a_\rho P_{-j} = 0, \text{ any } p_j \text{ in } [0, 1] \text{ is a local completion.} \end{aligned}$$

At a positive relevance probability, $x_j = 0$ still leads to yes by the equality rule. When relevance has probability zero, no universal yes completion is imposed.

14.1.3.3 Proposition 1 (complete fixed-proposal ballot PBE set) Let $Z = \{j : p_j = 0\}$. The complete independent-behavioral ballot correspondence is exactly the union of the following two classes.

1. **Positive-product class.** Z is empty, so $P_W > 0$. The hegemon uses the threshold $h_\theta = 1\{y \geq o_\theta\}$. If $a_\rho > 0$, every $p_j = 1$. If $a_\rho = 0$, each p_j may be any number in $(0, 1]$ independently.

2. **Coordinated-failure class.** $|Z| \geq 2$. Then $P_W=0$, both types of H vote yes, and all positive components of p may be arbitrary in $(0,1]$.

A profile with exactly one zero component is never an equilibrium. With that single voter held at no, every other weak action may be locally irrelevant, but the lone no voter faces positive relevance because all other weak voters and both types of H vote yes; equality or a positive offer makes it vote yes.

Proof. Lemma 1 pins down H . If $P_W=0$, it makes both types vote yes. A lone zero then violates Lemma 2; two or more zeros make every weak voter's relevance probability zero and hence satisfy it. If $P_W>0$, Lemma 1 gives the type thresholds. Lemma 2 forces all weak probabilities to one whenever the posterior-weighted hegemon acceptance probability is positive, and otherwise leaves every strictly positive probability as a completion. These cases exhaust $[0,1]^n$.

The condition $a_{\rho}=0$ in the positive-product class occurs precisely when:

- $y < o_0$ (both types of H vote no), for any ρ ; or
- $o_0 \leq y < o_1$ and $\rho=1$ (only the off-support low type votes yes).

The coordinated-failure class exists if and only if $n \geq 2$, equivalently $N \geq 4$. This is the sole population split needed at the ballot stage.

14.1.3.4 Pure profiles and mixed completions The pure ballot profiles are particularly transparent:

- all weak nonproposers vote yes and H follows the direct type threshold; or
- when $N \geq 4$, at least two weak nonproposers vote no and both types of H vote yes.

The mixed positive-product completions in Proposition 1 are substantive for the full assessment when a zero-probability ballot belief puts probability one on the high type. If the true preproposal belief assigns positive probability to the low type, its proposal may pass with probability P_W even though the weak voters' off-path belief makes their individual relevance zero. Therefore these probabilities may not be quotiented out before proposal optimality is evaluated.

14.1.4 Full outcome and payoff vector

For any ballot element (p, h_0, h_1) in Proposition 1 and each type θ , the conditional terminal distribution is

$$\begin{aligned} \Pr(\text{pass with } H \mid \theta) &= h_{\theta} P_W, \\ \Pr(\text{weak-caused failure} \mid \theta) &= h_{\theta} (1-P_W), \\ \Pr(\text{quota-impossible opt-out} \mid \theta) &= 1-h_{\theta}. \end{aligned}$$

The complete type-conditional expected payoff vector is

$$\begin{aligned} u_i(\theta) &= h_{\theta} P_W r_i, \\ u_j(\theta) &= h_{\theta} P_W x_j \quad \text{for each } j \neq i, \\ u_H(\theta) &= o_{\theta} + h_{\theta} P_W (y - o_{\theta}). \end{aligned}$$

The proposal-stage expected payoff of the recognized proposer and H are

$$\begin{aligned} v_i(s, b; \nu) &= (1-\nu)u_i(0) + \nu u_i(1), \\ \bar{h}(s, b; \nu) &= (1-\nu)u_H(0) + \nu u_H(1). \end{aligned}$$

These formulas preserve every named weak payoff and both type-specific hegemon payoffs. They also preserve the opt-out/inclusion distribution above; monetarily equal failure branches are not collapsed.

14.1.5 Proposal optimality as an assessment-level object

A recognized-proposer continuation assessment consists of:

1. a proposal distribution σ_i ;
2. a ballot belief $\rho_i(s)$ at every proposal, equal to ν on the support of σ_i and unrestricted by Bayes at a globally off-path proposal;
3. one ballot element $b_i(s)$ from Proposition 1 at every proposal;
4. support proposals that maximize $v_i(s, b_i(s); \nu)$; and

5. after that maximization, support only among maximizers minimizing $\bar{h}(s, b_i(s); \nu)$. If both quantities tie, different payment vectors, ballot distributions, and terminal outcomes remain in the correspondence.

This parameterization is an exact characterization of all proposal-stage PBEs. The next two propositions give closed-form implications without discarding the assessment-level correspondence.

14.1.6 The special case $N=3$

With $N=3$, there is one weak nonproposer. The coordinated-failure class is unavailable. At an on-path proposal:

- if $y < o_0$, both types of H vote no; the weak yes probability can be any number in $(0, 1]$, and the proposal fails;
- if $o_0 \leq y < o_1$ and $\nu < 1$, the weak voter votes yes, the low type accepts, and the high type opts out;
- if $o_0 \leq y < o_1$ and $\nu = 1$, the weak yes probability can be any number in $(0, 1]$; the actual high type opts out, while the counterfactual low-type payoff retains that probability;
- if $y \geq o_1$, both types and the weak voter vote yes.

Define

$$G = 1 - o_1,$$

$$L(\nu) = (1 - \nu)(1 - o_0).$$

G is the proposer payoff forced by the deviation $(y = o_1, x = 0)$: with a single weak nonproposer, its zero offer is relevant and equality forces yes. A separating off-path proposal can instead be assigned $\rho = 1$ and any strictly positive weak yes probability. Its deviation payoff can be made arbitrarily small, but not exactly zero when $\nu < 1$ and its residual is positive.

14.1.6.1 Proposition 2 (recognized-proposer PBEs for $N=3$) If $\nu < 1$:

1. When $G > 0$, the canonical pooling proposal $(y = o_1, x = 0)$ is always a PBE proposal outcome and gives the proposer G .
2. A separating PBE proposal outcome exists for every value V satisfying

$$0 < V \leq L(\nu), \quad V \geq G.$$

It has

$$r_i = V / (1 - \nu),$$

$$o_0 \leq y < o_1,$$

$$y + r_i \leq 1,$$

$$x = 1 - y - r_i,$$

all weak voters yes, low-type passage, and high-type opt-out. At $V = G > 0$, the forced pooling deviation ties the proposer but pays more to H, so the proposal tie-break selects the separating proposal. If $L(\nu) < G$, this separating set is empty.

3. When $G = 0$ (equivalently $o_1 = 1$), the canonical pooling proposal is not a PBE for $\nu < 1$: some positive-residual separating deviation has a strictly positive payoff under every admissible positive-product completion. The separating payoff set is the nonclosed interval $(0, L(\nu)]$.

The canonical pooling outcome in item 1 can coexist across assessments with separating outcomes whose value exceeds or equals G . In the pooling assessment, every off-path separating proposal is assigned $\rho = 1$ and a small enough positive weak yes probability to give strictly less than G . In a separating assessment with $V = G$, the pooling proposal is an unavoidable tie and the lower expected payment to H selects separation. This asymmetry is a consequence of proposal-contingent off-path completions, not an additional selection rule.

If $\nu = 1$:

1. when $G > 0$, the canonical pooling proposal is the unique on-path monetary outcome and gives the proposer G , the other weak state zero, and the high type of H o_1 ;

2. when $G=0$, every fixed-proposal ballot equilibrium in Proposition 1 is a proposal-stage PBE outcome. All actual weak payoffs are zero and the actual high type receives $o_1=1$; proposal mixtures and counterfactual low-type continuations remain part of the correspondence.

For $\nu < 1$, nontrivial proposal mixing does not add an outcome class. A pooling and separating proposal with the same proposer payoff give strictly different expected payoffs to H , so the proposal tie-break retains only the lower one if both are maximizers in the same assessment. Two separating support proposals with the same proposer and H payoffs have the same y , residual, and (because there is one recipient) transfer.

Proof. The fixed-proposal outcomes follow from Proposition 1 with $n=1$. Pooling deviations cannot be suppressed and have maximum residual G . At an off-path separating proposal with residual $r > 0$, choosing $\rho=1$ and p in $(0, 1]$ produces deviation payoff $(1-\nu)p$; for any target $V > 0$, p can be chosen small enough to keep this payoff below V . It cannot be made zero if $\nu < 1$. These facts yield the necessity and constructions above. The stated comparison of expected hegemon payoffs applies the only authorized proposal tie-break.

14.1.7 The general case $N \geq 4$

At every feasible proposal, including every deviation, at least two weak nonproposers can be assigned probability zero of yes. Proposition 1 then makes both types of H vote yes and gives every weak voter zero relevance. This coordination-failure completion gives the proposer zero, all weak states zero, and type θ of H its outside payoff. Thus every proposal deviation can be held to zero without changing the primitive action space or deleting any gift.

Let again

$$\begin{aligned} G &= 1 - o_1, \\ L(\nu) &= (1-\nu)(1-o_0), \\ M(\nu) &= \max\{G, L(\nu)\}. \end{aligned}$$

14.1.7.1 Proposition 3 (recognized-proposer PBEs for $N \geq 4$) Every proposer payoff V in $[0, M(\nu)]$ occurs in a PBE, but this scalar range does not replace the payoff-vector correspondence.

For $V > 0$, every pure on-path outcome belongs to one of these passage classes:

1. **Separating passage** (available only if $\nu < 1$):

$$\begin{aligned} 0 &< V \leq L(\nu), \\ r_i &= V / (1-\nu), \\ o_0 &\leq y < o_1, \\ y + r_i &\leq 1, \\ \sum_{j \neq i} x_j &= 1 - y - r_i. \end{aligned}$$

Every weak nonproposer votes yes, the low type accepts, and the high type opts out. Conditional payoffs are $(r_i, (x_j), y)$ for type zero and $(0, (o_j), o_1)$ for type one.

2. **Pooling passage:**

$$\begin{aligned} 0 &< V \leq G, \\ r_i &= V, \\ o_1 &\leq y \leq \min\{y_{\bar{}}, 1-V\}, \\ \sum_{j \neq i} x_j &= 1 - y - V. \end{aligned}$$

Every voter and both types of H vote yes. Conditional payoffs are $(V, (x_j), y)$ for both types.

Any one such positive-payoff proposal is made a strict proposal optimum by assigning coordinated failure to every off-support proposal. Gifts and oversized unanimous support therefore survive whenever they are embedded in the chosen feasible vector; no minimum coalition conclusion follows.

For $V=0$, a fixed-proposal ballot outcome is proposal-stage admissible if and only if

$$\begin{aligned} v_i(s, b; \nu) &= 0 \\ \text{and} \\ hbar(s, b; \nu) &= (1-\nu)o_0 + \nu o_1. \end{aligned}$$

The second equality is required by the proposal tie-break because an off-support coordinated-failure proposal necessarily ties the proposer at zero and pays H exactly its expected outside option. This zero class contains:

- every terminal-failure ballot completion;
- if $\text{nu} < 1$, a separating passage outcome only when $\text{r}_i = 0$ and $\text{y} = \text{o}_0$;
- if $\text{nu} = 1$, every separating positive-product completion, and a pooling passage outcome exactly when $\text{y} = \text{o}_1$ and $\text{r}_i = 0$ (arbitrary feasible named gifts may consume $1 - \text{o}_1$).

Proposal mixtures are possible only over support elements with the same proposer payoff and the same minimal expected payoff to H . Within a positive separating support this fixes y , the residual, and the total gifts but may leave their distribution across named recipients free. The analogous result holds within a pooling support. At zero, mixtures over different failure or passage outcomes remain whenever both displayed equalities hold. These joint distributions are retained rather than quotiented by player-by-player local irrelevance.

Proof. Proposition 1 supplies coordinated failure after every proposal, so a chosen positive payoff is strict against all deviations. The two passage classes and their feasibility bounds follow directly from the full payoff formula. Conversely, positive proposer payoff requires passage with positive probability. At an on-path belief $\text{nu} < 1$, any posterior-supported accepting type makes every weak voter choose yes, so separating or pooling passage is deterministic conditional on the accepting types; at $\text{nu} = 1$, only pooling can give the actual proposer a positive payoff. These are exactly the two listed classes. At payoff zero, an unavoidable coordinated-failure deviation pins down the tie-break benchmark for H , yielding the two displayed equalities and the concrete subclasses.

14.1.8 Pre-recognition continuation interface

The object imported by Round 1 is pre-recognition and integrates the uniform draw. A complete Round-2 assessment α selects, for every recognized weak proposer i , a proposal-stage PBE element $\mathbf{e}_i$ characterized above. For weak state k and type θ , write $u_k^i(\theta; \mathbf{e}_i)$ for the full payoff formula in Section 3. Then

$$\begin{aligned} C_2, U^\alpha(h_2)_k &= (1/m) \sum_i [(1-\text{nu})u_k^i(0; \mathbf{e}_i) + \text{nu} u_k^i(1; \mathbf{e}_i)], \\ C_2, U^\alpha(h_2)_H(\theta) &= (1/m) \sum_i u_H^i(\theta; \mathbf{e}_i). \end{aligned}$$

The interface also carries the corresponding uniform mixture of the three terminal-outcome probabilities for each type. The assessment may be identity-asymmetric: isomorphic proposer nodes can select different elements of the same correspondence. Symmetry is available but is not imposed.

This averaging is the only operation performed at pre-recognition. No β is used in Round 2. A Round-1 predecessor must multiply the selected frozen Round-2 payoff vector by β exactly once where its own transition requires that continuation.

14.1.9 Beliefs, existence, multiplicity, and boundaries

- **Bayes on path.** Every support proposal has ballot belief $\rho = \text{nu}$.
- **Off path.** Every zero-probability proposal records an explicit ρ and a full ballot element from Proposition 1. The payoff from a proposal deviation is still integrated using nu .
- **After the ballot.** The game is terminal. The public vector can reveal H 's type, but no successor belief enters a payoff or action.
- **Existence.** A PBE exists for every $N \geq 3$, nu in $[0, 1]$, and $0 \leq \text{o}_0 < \text{o}_1 \leq \text{y}_{\text{bar}} \leq 1$.
- **Multiplicity.** Off-path completions are generally multiple. On-path payoff multiplicity is exactly as described in Propositions 2 and 3.
- $\text{o}_0 = 0$. The region $\text{y} < \text{o}_0$ is empty; all other formulas remain valid.
- $\text{o}_1 = 1$. $G = 0$. For $N = 3, \text{nu} < 1$, the proposer payoff set is nonclosed at zero; for $N \geq 4$, coordinated failure restores the zero endpoint.
- $\text{nu} = 0$. Proposition 2 applies through its $\text{nu} < 1$ branch; separating and pooling are both possible across assessments. Proposition 3 is unchanged.
- $\text{nu} = 1$. Counterfactual low-type strategies and payoffs remain recorded even though proposal optimality uses only the actual high type.

- **y_bar=o_1.** The pooling rent interval above o_1 disappears, but the canonical pooling proposal remains feasible. Separating feasibility is unaffected because $y < o_1 \leq y_bar$.
- **Equality.** $y = o_theta$ makes type **theta** vote yes. A relevant weak voter offered zero votes yes. No additional threshold convention is used.

14.1.10 Claims ledger

- **Exact fixed-proposal independent ballot correspondence.**
 - **Method:** analytic fixed-point proof
 - **Status:** proved
- **H threshold when all weak votes have positive probability.**
 - **Method:** direct expected-payoff difference
 - **Status:** proved
- **H yes under sure weak failure.**
 - **Method:** outcome-signature relevance plus equality rule
 - **Status:** proved
- **N=3 recognized-proposer correspondence.**
 - **Method:** analytic deviation construction and necessity
 - **Status:** proved
- **N>=4 recognized-proposer correspondence.**
 - **Method:** analytic coordinated-failure construction and necessity
 - **Status:** proved
- **Full type/player payoff and outcome vectors.**
 - **Method:** direct terminal accounting
 - **Status:** proved
- **Uniform pre-recognition integration.**
 - **Method:** law of total expectation
 - **Status:** proved
- **Pure-ballot enumeration and boundary checks.**
 - **Method:** node-specific R verifier
 - **Status:** checked numerically
- **Independent formal/adversarial acceptance.**
 - **Method:** two exact-hash read-only reviews
 - **Status:** proved by review bundle
- **Inclusion in standalone output.**
 - **Method:** post-freeze render
 - **Status:** checked separately

14.1.11 Invalidation rule

Any change to the frozen Gate 0 bundle, proposal budget, outcome signature, equality convention, simultaneous-information structure, belief discipline, or opt-out/failure payoff reopens this node. Every Round-1 or later artifact that imports this interface must then be invalidated and rederived from the new content hash.

14.2 Round 2, majority, active H: repaired terminal derivation

State. r2_majority_active_h

Dependency. Frozen Gate 0 bundle sha256:6e28cb3f...

Solution concept. PBE, with the Gate 0 response-at-equality rule

Payoff date. Round 2 native units; no **beta** occurs here

Post-freeze status. Repaired exact interface accepted by two independent read-only rereviews.

This note rederives the terminal majority node from the frozen primitives. It does not assume a minimal coalition, zero gifts, or common ballot completions across proposals. The repair separates the public posterior before proposal, ν , from the belief at the ballot following proposal s , $\rho(s)$.

14.2.1 State, beliefs, and notation

There are $N=m+1 \geq 3$ states: the hegemon H and m weak states. Weak state i is recognized uniformly to propose. The majority quota remains

$$q = \text{floor}(N/2)+1$$

even after an H no. The proposer is counted as yes. The simultaneous ballot therefore contains H and $n=N-2$ weak nonproposers. A feasible proposal is

$$\begin{aligned} s_i &= (y, (x_j)_{j \neq i}), \\ 0 &\leq y \leq y_{\text{bar}}, \\ x_j &\geq 0, \\ X &= \sum_{j \neq i} x_j \leq 1-y, \\ r_i &= 1-y-X. \end{aligned}$$

If passage includes H , the proposer receives r_i . If H votes no and the weak votes alone meet q , H opts out, y is reabsorbed, and the proposer receives $1-X$. A named x_j is paid after passage even if j voted no.

Let ν be the true public posterior at the full Round-2 history before the proposal. Recognition and the proposer strategy are type-independent. Hence

$$\rho(s) = \nu$$

at every proposal in the proposer strategy's support. At a globally off-path proposal, PBE permits any explicitly stated $\rho(s)$ in $[0, 1]$. This freedom changes the voters' best responses; it does not change Nature's distribution. The recognized proposer evaluates a deviation under ν , not under $\rho(s)$.

For weak voting probabilities $p = (p_j)$, define

$$\begin{aligned} a &= q-3, & b &= q-2=a+1, \\ K &= \sum_j Y_j, & K_{-j} &= \sum_{k \neq j} Y_k, \\ A(p) &= \Pr_p(K \geq b), & B(p) &= \Pr_p(K \geq b+1), \\ S(p) &= \#\{j : p_j = 1\}, & L(p) &= \#\{j : p_j > 0\}. \end{aligned}$$

All behavioral randomizations are independent conditional on the ballot information set. Impossible count events have probability zero.

14.2.2 Terminal transition and complete payoff map

The four Gate 0 branches are:

- **yes.**
 - **Weak-nonproposer count:** $K \geq b$
 - **Outcome:** PR11, pass with H
 - **Payoffs:** $H: y$, proposer r_i , named $j: x_j$
- **yes.**
 - **Weak-nonproposer count:** $K \leq a$
 - **Outcome:** PR12, active- H failure
 - **Payoffs:** $H: o_{\text{theta}}$, all weak zero
- **no.**
 - **Weak-nonproposer count:** $K \geq b+1$
 - **Outcome:** PR13, weak-only pass after opt-out
 - **Payoffs:** $H: o_{\text{theta}}$, proposer $1-X$, named $j: x_j$
- **no.**
 - **Weak-nonproposer count:** $K \leq b$
 - **Outcome:** PR14, failure after opt-out
 - **Payoffs:** $H: o_{\text{theta}}$, all weak zero

For any fixed-proposal ballot profile and each type theta , write h_{theta} for H 's yes action. The exact terminal probabilities are

$$\begin{aligned} \Pr(\text{PR11}|\theta) &= h_\theta A, \\ \Pr(\text{PR12}|\theta) &= h_\theta (1-A), \\ \Pr(\text{PR13}|\theta) &= (1-h_\theta)B, \\ \Pr(\text{PR14}|\theta) &= (1-h_\theta)(1-B). \end{aligned}$$

Consequently,

$$\begin{aligned} u_H(\theta) &= o_\theta + h_\theta A(y - o_\theta), \\ u_i(\theta) &= h_\theta A r_i + (1-h_\theta)B(1-X), \\ u_j(\theta) &= [h_\theta A + (1-h_\theta)B]x_j. \end{aligned}$$

These formulas retain type, player identity, inclusion, opt-out, terminal branch, and implemented-payment coordinates.

14.2.3 The ballot game at fixed (s, ρ)

This section is deliberately solved before proposal optimality.

14.2.3.1 Lemma 1: H's response If H votes yes, agreement includes it exactly when $K \geq b$; otherwise it receives o_θ . A no always gives o_θ . Thus

$$\Delta_H(\theta) = A(p)(y - o_\theta).$$

Changing H's ballot changes at least opt-out status at every weak vote profile, so its relevance probability is one even when $A=0$. The Gate 0 equality rule therefore gives the unique prescribed action

$$h_\theta = 1\{A=0 \text{ or } y \geq o_\theta\}.$$

In particular, H votes yes at $y = o_\theta$ and both types vote yes when $A=0$.

14.2.3.2 Lemma 2: a weak nonproposer's response If H votes yes, weak voter j toggles passage when $K_j = a$. If H votes no, it toggles weak-only passage when $K_j = b$. Let

$$t_\rho(y) = (1-\rho)h_0 + \rho h_1.$$

The exact relevance probability and monetary payoff difference are

$$\begin{aligned} R_j &= t_\rho(y)\Pr(K_j = a) + (1-t_\rho(y))\Pr(K_j = b), \\ \Delta_j &= x_j R_j. \end{aligned}$$

Every other profile contributes zero. Hence $p_j = 1$ when $R_j > 0$, including at $x_j = 0$, while any p_j in $[0, 1]$ is an explicit completion when $R_j = 0$.

14.2.3.3 Proposition 1: exhaustive fixed-proposal ballot correspondence Every independent-behavioral ballot equilibrium at (s, ρ) belongs to exactly one of two classes.

F. Sure active-H failure. This class exists iff $q \geq 4$, equivalently $N \geq 6$, and is exactly

$$L(p) \leq q-4.$$

It has $A=0$, both H types yes, sure PR12, type- θ H payoff o_θ , and zero weak payoffs. At most $q-4$ weak voters may use arbitrary positive yes probabilities; every other weak voter uses zero. The class is independent of ρ .

S. Secured passage under the ballot belief. This class exists at every proposal and has $A=1$, $h_\theta = 1\{y \geq o_\theta\}$, and

$$\begin{aligned} \text{if } t_\rho(y) = 1: & S(p) \geq q-2, \\ \text{if } t_\rho(y) < 1: & S(p) \geq q-1. \end{aligned}$$

Every remaining weak voter may use any independent probability in $[0, 1]$. The first condition allows $B(p)$ anywhere in $[0, 1]$; the second forces $B(p) = 1$.

Proof of exhaustiveness. If $A=0$, independence gives $L \leq b-1$. At $L=b-1$, choose a zero-probability voter. With positive probability all L positive voters vote yes, putting $K_{-j}=b-1=a$; both H types vote yes, so this voter is relevant and must vote yes, a contradiction. Conversely, $L \leq b-2=q-4$ removes both pivotal counts from every weak voter's support.

Now suppose $A>0$. H follows the type threshold. If $t_{\rho}=1$ and fewer than b voters are sure yes, $A>0$ implies that some unsure voter can face exactly $b-1$ other yes votes with positive probability. It is relevant and must be sure yes, a contradiction. Thus $S \geq b$; this also suffices. If $t_{\rho}<1$ and $S \leq b$, either a zero voter or a positive unsure voter can face exactly b other yes votes with positive probability. The H -no branch has positive probability, so this voter must be sure yes, again a contradiction. Thus $S \geq b+1$; this also suffices. These sure-vote bounds make $A=1$, proving that no equilibrium has $0 < A < 1$.

For a separating proposal $o_0 \leq y < o_1$, $t_{\rho}=1-\rho$. Therefore $\rho=0$ allows exactly $q-2$ sure weak votes and any B in $[0,1]$, while every $\rho>0$ forces at least $q-1$ sure weak votes and $B=1$. This discontinuity is the reason ρ cannot be replaced by ν off path.

14.2.4 Fixed-proposal payoffs under the true ν

The ballot in Proposition 1 is rational under ρ . Proposal optimality uses

$$g_k(s, b; \nu) = (1-\nu)u_k(0) + \nu u_k(1).$$

For class F ,

$$\begin{aligned} g_i &= 0, \\ g_j &= 0 \text{ for every weak } j, \\ g_H &= o_{\bar{b}}(\nu) = (1-\nu)o_0 + \nu o_1. \end{aligned}$$

For class S , define $t_{\nu}(y) = (1-\nu)h_0 + \nu h_1$. Then

$$\begin{aligned} g_i &= [t_{\nu} + (1-t_{\nu})B](1-X) - t_{\nu} y, \\ g_j &= [t_{\nu} + (1-t_{\nu})B]x_j, \\ g_H &= (1-\nu)[h_0 y + (1-h_0)o_0] \\ &\quad + \nu[h_1 y + (1-h_1)o_1]. \end{aligned}$$

At an on-path proposal, $\rho=\nu$. If $t_{\nu}<1$, Proposition 1 forces $B=1$; if $t_{\nu}=1$, all types in the support of ν accept. In both cases the on-path proposer payoff simplifies to

$$g_i = 1-X-t_{\nu}(y)y.$$

That simplification is generally false off path. For example, when $o_0=0 \leq y < o_1$, setting $\rho=0$ and exactly $q-2$ sure weak yes votes permits $B=0$. The true- ν deviation payoff is then only

$$(1-\nu)r_i,$$

even though the weak voters act under certainty of the low type.

At any positive-probability terminal history, Bayes' rule starts from $\rho(s)$, not from ν unless the proposal is on path:

$$\begin{aligned} \Pr(\theta=1 | H \text{ action } d, s) \\ = \rho(s) 1\{h_1=d\} / [(1-\rho(s)) 1\{h_0=d\} + \rho(s) 1\{h_1=d\}]. \end{aligned}$$

The weak vote vector adds no type likelihood conditional on the proposal. Zero-probability terminal beliefs are stated, but no later player moves.

14.2.5 Exact proposal-stage assessment

A continuation assessment for recognized proposer i contains:

1. a proposal distribution σ_i ;
2. a belief map $\rho_i(s)$, equal to ν on support and explicitly stated at every globally off-path proposal;
3. a class- F or class- S ballot element from Proposition 1 at each proposal;

4. the resulting type-conditional terminal distribution, payoffs, and terminal beliefs;
5. support only on maximizers of $g_i(s, b_i(s); \nu)$; and
6. among those maximizers, support only on proposals minimizing $g_H(s, b_i(s); \nu)$.

If proposer and H expected payoffs tie, differences in gifts, identities, type-contingent outcomes, ballot completions, and terminal signatures remain in the correspondence. This functional description is the exact proposal-stage PBE characterization. The following propositions derive its scalar projections and constructive support restrictions.

14.2.6 Proposal-stage implications for $3 \leq N \leq 5$

Here class F is unavailable, so every proposal receives a class-S completion.

14.2.6.1 Proposition 2: $\nu_0 > 0$ The deviation $y=0, X=0$ has both H types vote no under every ρ . Because $t_\rho=0 < 1$, class S forces $B=1$, so this deviation gives the proposer one. No proposal can give more than one. Hence the proposer payoff is uniquely one.

The complete tie-refined on-path condition is

$$X=0 \text{ and } t_\nu(y)y=0.$$

These conditions also give H exactly $\bar{o}(\nu)$, the lower bound supplied by its type-specific optimality. Thus all such on-path proposals survive the tie-break. The supported-type payoff vector gives the proposer one and every other weak state zero, but proposal, inclusion, opt-out, off-support type outcomes, and ballot completions may differ and are retained.

14.2.6.2 Proposition 3: $\nu_0=0$ Define

$$\begin{aligned} G &= 1 - \nu_1, \\ L &= 1 - \nu, \\ D &= \max\{G, L\}. \end{aligned}$$

The pooling deviation $y=\nu_1, X=0$ cannot be suppressed: both H types accept under every ρ , so it gives G . At the deviation $y=0, X=0$, the low type accepts and the high type rejects. Every ballot completion gives at least L ; the minimum L is attained with $\rho=0$, exactly $q-2$ sure weak yes votes, and $B=0$. Thus every equilibrium proposer payoff satisfies $V \geq D$.

These are the only unavoidable lower bounds. Any separating deviation can be assigned $\rho=0$ and $B=0$, giving $(1-\nu)r_i \leq L$. Any pooling deviation gives at most G . Moreover every V in $[D, 1]$ is attained on path by $y=0, X=1-V$,

with $\rho=\nu$ and a class-S ballot completion. Therefore the exact proposer payoff projection is $[D, 1]$.

For $V > D$, every on-path class-S outcome with proposer payoff V can be made the unique payoff maximum by assigning all other proposals strictly less than V . At $V=D$, the primitive H-payoff tie-break adds the following exact restriction:

- if $L > G$, the unavoidable $y=0, X=0$ tie gives H $\bar{o}(\nu)$, so support outcomes must also give H $\bar{o}(\nu)$;
- if $G > L$, the unavoidable $y=\nu_1, X=0$ tie gives H ν_1 , so support outcomes must give H no more than ν_1 ;
- if $G=L$, both ties are unavoidable and the smaller H payoff is $\bar{o}(\nu)$, so support outcomes must give H $\bar{o}(\nu)$.

If several proposals are mixed, every support proposal must have both the same proposer payoff and the same tie-minimal H payoff. Gifts and identities are otherwise unrestricted subject to feasibility and the displayed conditions.

At the degenerate corner $\nu=1, \nu_1=1, D=0$. The two pure proposal forms with zero value are $y=0, X=1$ and $y=1, X=0$; both give the actual high type payoff one. Their payment and opt-out signatures remain distinct.

14.2.7 Proposal-stage implications for $N \geq 6$

Class F is available at every proposal and every ρ . Thus an assessment can assign zero payoff to every proposal deviation.

For every V in $(0,1]$, any on-path class-S outcome satisfying

$$V=1-X-t_{\nu}(y)y>0$$

is supportable by assigning F to every other proposal. Every V is attained, for example, by $y=0, X=1-V$. At value zero, class- F outcomes are supportable at arbitrary proposals. The H -payoff tie-break also permits a class- S outcome at value zero exactly when it gives H its outside expectation:

$$1-X-t_{\nu}(y)y=0, \\ g_H=o_{\bar{\nu}}(\nu).$$

Feasibility makes these two equations equivalent to either

$$y=0, X=1,$$

or

$$y>0, t_{\nu}(y)=1, X=1-y, \\ \text{and } o_{\theta}=y \text{ for every type with positive probability under } \nu.$$

Hence the proposer payoff projection is $[0,1]$. It is not the continuation interface. For example, a zero-value F outcome pays every weak state zero, whereas a zero-value S outcome may implement a unit of named weak gifts. Both survive the primitive tie-break and cannot be quotiented.

14.2.8 Pre-recognition continuation correspondence

For each weak identity i , let α_i be any recognized-proposer assessment satisfying Sections 5–7, including its proposal distribution, ρ map, ballot completions, type-conditional outcomes, and beliefs. Let $U^{\{i,\alpha_i\}}$ denote the full resulting vector. Before uniform recognition,

$$C2_M_active(h2) \\ = \{ (1/m) \sum_{i=1}^m U^{\{i,\alpha_i\}} : \\ \alpha_i \text{ is admissible at } \nu(h2) \}.$$

In particular,

$$C_{H,\theta}=(1/m)\sum_i U_{H,\theta}^{\{i,\alpha_i\}}, \\ C_{W,j,\theta}=(1/m)\sum_i U_{W,j,\theta}^{\{i,\alpha_i\}}.$$

The same averaging retains the PR11–PR14 terminal probabilities, inclusion and opt-out status, and implemented payment vectors. Full $h2$ remains an index even when two histories share ν , because their off-path ρ and completion maps may differ. Recognition is independent of type and history, so it does not update ν .

14.2.9 Boundaries, existence, and invalidation

- $N=3$: $q=2, b=0$; class F is impossible. Class S requires no sure weak yes at $t_{\rho}=1$ and the sole weak nonproposer sure yes at $t_{\rho}<1$.
- $N=4,5$: $q=3, b=1$; class F is impossible. Class S requires one sure weak yes at $t_{\rho}=1$ and two at $t_{\rho}<1$.
- $N \geq 6$: class F exists exactly at $L(p) \leq q-4$.
- $\rho=0$ at a separating proposal permits B in $[0,1]$; every $\rho>0$ forces $B=1$. This statement concerns ballot rationality.
- $\nu=0$ and $\nu=1$ govern the true proposal payoff. They never overwrite an explicitly different off-path ρ .
- Both H type strategies and both type-conditional payoff vectors are retained even at degenerate beliefs.
- $y=o_{\theta}$ makes H vote yes. Positive weak relevance makes a weak voter vote yes even at $x_j=0$.
- A PBE exists for every admissible primitive vector, population, posterior, and full public history.

The exact reviewed interface is `interfaces/r2_majority_active_h_v1.json`. Its repaired exact hash was accepted by two independent read-only rereviews. Any change to the frozen Gate 0 bundle invalidates this node and every descendant. An R1 importer must use the reviewed frozen replacement and apply `beta` exactly once; this R2 object uses no discount.

14.3 Round 2 majority after the hegemon's opt-out

14.3.1 Scope and frozen input

This note derives only the terminal node `r2_majority_weak_only`. Its frozen input is `gate0_bundle_v1.json` with SHA-256 `6e28cb3f...`. No historical equilibrium result is imported. The derivation is in native Round-2 units and applies no discount factor.

At the public history `h2`, the hegemon has already opted out irreversibly. Its outside payoff was realized in Round 1, so its current Round-2 flow is zero. There are $m=N-1$ weak states, the original majority quota remains

$$q = \text{floor}(N/2)+1,$$

and a uniformly recognized weak proposer is counted as voting yes. Therefore there are $K=m-1=N-2$ simultaneous nonproposer voters and the proposal passes if at least

$$t = q-1$$

of those K voters choose yes. For every $N \geq 3$, $1 \leq t \leq K$; in particular, the node is nonempty when $N=3$.

The recognized proposer i chooses nonnegative named gifts x_j for every $j \neq i$; the residual is

$$x_i = 1 - \sum_{j \neq i} x_j \geq 0.$$

Thus a passing proposal implements a vector x in the unit simplex Δ_m ; a failed proposal gives every weak state zero. A named gift is paid when the proposal passes even if its recipient voted no.

14.3.2 Terminal ballot correspondence

Fix a public proposal and a recognized proposer. Let p_j in $[0,1]$ be weak nonproposer j 's independent behavioral probability of yes. Write

- $a(p)$ for the number of voters with $p_j=1$;
- $c(p)$ for the number with $0 < p_j < 1$;
- $\text{ell}(p)=a(p)+c(p)$ for the number that can vote yes with positive probability.

For voter j , let S_{-j} be the number of yes votes among the other nonproposers. Its vote is relevant exactly on $S_{-j}=t-1$. Therefore

$$\rho_j(p_{-j}) = \Pr(S_{-j}=t-1)$$

is its relevance probability and its ordinary expected-payoff difference is

$$EU_j(\text{yes}) - EU_j(\text{no}) = x_j \rho_j(p_{-j}) \geq 0.$$

The frozen response convention adds the consequential equality case: if $\rho_j > 0$, yes is required even when $x_j=0$; if $\rho_j=0$, the action is an explicit completion and may be pure or mixed. Hence the exact ballot condition is

$$p_j < 1 \text{ implies } \rho_j(p_{-j})=0.$$

14.3.2.1 Proposition 1 (complete pure and behavioral ballot characterization) A behavioral profile p is a terminal ballot equilibrium if and only if

$$a(p) \geq t \text{ or } \text{ell}(p) \leq t-2. \quad (B)$$

The first class passes with probability one. The second class fails with probability one. Consequently there is no terminal ballot equilibrium with a nondegenerate probability of passage. For pure profiles, if a is the number of nonproposer yes votes, condition (B) becomes

$$a \geq t \text{ or } a \leq t-2;$$

the knife-edge count $a=t-1$ is never an equilibrium.

Proof. Independent interior Bernoulli variables put positive probability on every integer between the minimum and maximum possible yes counts. Suppose first that $a \geq t$. Every voter with $p_j < 1$ faces at least t certain yes votes among the other voters, so $S_{-j}=t-1$ has probability zero. Voters with $p_j=1$ satisfy the response condition. Passage is certain.

If $e_{11} \leq t-2$, even every voter who can vote yes doing so leaves fewer than t nonproposer yes votes. For an interior voter, the other voters can supply at most $e_{11}-1 \leq t-3$ yes votes; for a zero-probability voter they can supply at most $e_{11} \leq t-2$. Thus every voter with $p_j < 1$ has zero relevance probability, and failure is certain.

Conversely, suppose neither inequality in (B) holds: $a < t$ and $e_{11} > t-1$. If some voter has $p_j = 0$, the support of S_{-j} contains every integer from a through e_{11} , including $t-1$; that voter has positive relevance probability but does not choose yes. If no voter has probability zero, then at least one voter is interior. Excluding that voter, the support runs from a through $K-1$; because $a \leq t-1$ and $K \geq t$, it contains $t-1$. Again a voter with positive relevance probability fails to choose yes. Both cases contradict the response condition. This proves necessity. QED.

The two classes in (B) are full, product-safe completions: every action profile in the support of a passing completion passes, and every action profile in the support of a failing completion fails. This does **not** identify the two classes with each other. Switching several actions can cross from one class to the other even though each switch is locally irrelevant against the original profile.

14.3.2.2 Small-N boundaries

- $N=3$: $K=t=1$. The unique ballot completion is yes and passage is certain at every proposal.
- $N=4$: $K=t=2$. The only behavioral completions are all-yes passage and all-no failure. At all no, either voter is locally irrelevant holding the other's no fixed, but changing both votes produces passage.
- $N=5$: $K=3$, $t=2$. Pure passage requires at least two certain yes votes; pure failure requires all no. With two certain yes votes, the third voter may use any behavioral probability because passage is already secured.

14.3.3 Proposal-contingent completions and proposer optimality

For recognized proposer i , let $\kappa_i(s)$ assign a full ballot completion to every feasible proposal s . Define $d_i(s)=1$ if that completion belongs to the certain-passage class of Proposition 1 and $d_i(s)=0$ if it belongs to the certain-failure class. The proposer's payoff from s is exactly

$$g_i(s) = d_i(s) \cdot x_i(s).$$

14.3.3.1 Proposition 2 (complete strategy characterization at a recognized node) A continuation assessment at the recognized-proposer node is a PBE if and only if:

1. at every proposal s , $\kappa_i(s)$ satisfies (B);
2. the completion map and proposal strategy are measurable;
3. $\arg\max_s g_i(s)$ is nonempty; and
4. the proposer's behavioral strategy is supported on $\arg\max_s g_i(s)$.

No additional belief choice enters this terminal weak-only ballot. The inherited public belief must satisfy the upstream Gate 0/Bayes requirements, but it is payoff- and action-irrelevant here because the hegemon is absent and no weak player observes its type.

Proof. Proposition 1 is necessary and sufficient for sequential rationality at every simultaneous ballot information set. Conditional on those ballot strategies, the proposer directly solves $\max_s g_i(s)$; support on its maximizers is necessary and sufficient for its sequential rationality. There is no later decision and no continuation value. QED.

For $N=3$, every proposal necessarily passes. The proposer uniquely chooses zero gift and obtains one, so the implemented vector is the unit vector e_i .

For every $N \geq 4$, both all yes and all no satisfy Proposition 1 at every proposal. Thus proposal-contingent completions can make any selected proposal pass and every distinct deviation fail. This yields both the all-failure PBE and passing PBEs for every allocation in Δ_m . More generally, Proposition 2 retains completion maps with several passing proposals and requires the proposer to choose only those with maximal residual.

The project tie-break does not reduce this set. The hegemon's current payoff is zero (and its already-realized outside payoff is the same) at every proposal. Hence every proposer-payoff tie also ties the hegemon payoff; when outcomes otherwise differ, the frozen contract retains the correspondence.

14.3.4 Exact payoff correspondences by identity

Let $\hat{v}_i$ be the expected weak payoff vector conditional on recognizing proposer i .

14.3.4.1 Pure strategies For $N=3$, $v_i=e_i$. For $N \geq 4$, the exact conditional pure-strategy payoff set is

$$S_i^{\text{pure}} = \{0_m\} \cup \Delta_m. \quad (P)$$

The zero vector is implemented by a failing proposal; any vector in the simplex is implemented by a passing proposal with that allocation.

14.3.4.2 Behavioral strategies For $N \geq 4$, the exact conditional behavioral payoff set is

$$S_i = \Delta_m \cup \{v \in R_+^m : v_i=0 \text{ and } \sum_j v_j \leq 1\}. \quad (M)$$

To prove necessity, let $M_i = \max_s g_i(s)$. If $M_i > 0$, every proposal in the proposer's support passes and pays the proposer M_i ; hence $v_i = M_i > 0$ and the expected allocation sums to one. If $M_i = 0$, then $v_i = 0$; mixing between passing residual-zero proposals and failing proposals can yield total expected weak surplus anywhere in $[0, 1]$.

For sufficiency, any $v \in \Delta_m$ is supported by making its corresponding proposal pass and all distinct proposals fail. If $v_i = 0$ and $T = \sum_j v_j < 1$, mix with probability T on the passing allocation v/T and with probability $1-T$ on a distinct failing proposal; both give the proposer zero. The cases $T=0$ and $T=1$ follow directly. This proves (M).

Before recognition, the uniformly integrated continuation interface is

$$C2_M_W0(h2) = \{ (0_H, w_1, \dots, w_m) :$$

$$w = (1/m) \sum_{i=1}^m v_i, v_i \in S_i \} \quad (C)$$

for behavioral PBEs. For pure PBEs, replace S_i by S_i^{pure} . The first coordinate in (C) is the hegemon's **current Round-2 flow**, not a second payment of its Round-1 outside option. The formula is indexed by the full public $h2$; its value correspondence is the same at every such history, while the assessment-level completion maps may differ across histories.

Every vector in (C) is nonnegative and has aggregate weak value between zero and one. Under pure proposal strategies the aggregate belongs to $\{0, 1/m, \dots, 1\}$. Behavioral proposal mixing makes every aggregate in $[0, 1]$ attainable: for each recognized proposer, mix with the desired probability on a residual-zero passing allocation to a different weak state and otherwise choose a failing proposal.

No equilibrium selection is supplied by the game. Downstream nodes must import the full correspondence (C), or a separately proved selection-free bound, and must not silently replace it with a scalar.

14.3.5 Gifts, oversized support, and the Gate 0 counterexample

Positive outsider gifts and oversized support are genuine equilibrium possibilities. For $N=5$, $q=3$, give each of the three nonproposers 0.2 and leave the proposer 0.4. Pair this proposal with two nonproposer yes votes and one no vote. Passage is already secured, so the no voter is locally irrelevant and receives a strictly positive outsider gift. Alternatively, pair the same proposal with all three yes votes; the third supporting vote is oversized. Pair every distinct proposal, including the zero-gift deviation, with all no. The on-path proposal gives the proposer 0.4, while every deviation fails and gives zero. All ballot actions satisfy Proposition 1. The same construction works for every strictly positive gift vector with positive proposer residual, a relatively open subset of the feasible simplex. A positive gift to a non-supporting weak voter and oversized support require $K > t$, which holds for every $N \geq 5$. For $N=4$, positive gifts can still be sustained, but every nonproposer vote is required for passage.

At all no, each isolated change to yes is locally irrelevant. Yet changing enough no votes jointly causes passage. It is therefore invalid to delete outsider gifts, reduce support to a minimal winning set, or quotient all locally irrelevant actions player by player. Such a representation would discard PBEs in (P)–(C).

14.3.6 Status and invalidation

- Proposition 1: **proved** and exhaustively checked for pure profiles for $3 \leq N \leq 15$, and on a five-point probability grid for $3 \leq N \leq 9$.
- Proposition 2: **proved**.
- Conditional and pre-recognition payoff correspondences: **proved**.
- $N=5$ gift/oversized-support construction: **proved** and checked numerically.
- Post-freeze interface status: **PASS after two independent read-only reviews**.

Any change to the frozen Gate 0 bundle, the original quota after opt-out, the equality response convention, independent behavioral randomization, named gift payment, or residual budget invalidates this entire node and every descendant that imports it.

15 Round 1: literal continuation import

The two Round-1 nodes import the frozen terminal elements by full public history, type, identity, payoff vector, outcome distribution, belief, and completion. A terminal payoff is multiplied by **beta** once, and only when it enters a Round-1 continuation branch.

15.1 Round 1 under unanimity

15.1.1 Status and frozen dependencies

- Node: **r1_unanimity**.
- Solution concept: Perfect Bayesian equilibrium (PBE).
- Native payoff date: Round 1.
- Gate 0 bundle: **sha256:6e28cb3f....**
- Frozen Round-2 batch: **sha256:00f84d42....**
- Frozen active-H unanimity continuation: **sha256:f3ca4ebf....**
- Post-freeze status: repaired exact interface accepted by two independent read-only rereviews.
- Rejected predecessor: **sha256:da52b135....**
- Machine-readable validated export domain: $N \geq 3$.
- $N=3$ status: **proved_consumable_after_independent_rereview**.

This node consumes the exact Round-2 correspondence. It does not rederive a Round-2 proposal, ballot, belief, or payoff. It also does not choose an element of that correspondence while checking a local Round-1 deviation. Instead, a complete Round-1 assessment states, in advance, one literal Round-2 element after every full failed vote vector. This is the selection that every current voter anticipates.

No result from a historical equilibrium classification is used below.

15.1.2 State, proposal, and simultaneous ballot

There are $N=m+1 \geq 3$ players. The active hegemon is **H**; the other m players are weak states. At full public history **h1**, the public belief is

$$\mu = \Pr(\theta=1 \mid h1).$$

One weak state **i** is recognized uniformly and independently of the type and history. It proposes

$$\begin{aligned} s_i &= (y, (x_j)_{j \in J_i}), \\ 0 &\leq y \leq y_{\text{bar}}, \\ x_j &\geq 0, \\ r_i &= 1-y-\sum_j x_j \geq 0, \end{aligned}$$

where J_i contains the $n=N-2$ weak nonproposers. The proposer is counted yes. All voters in J_i and **H** then vote simultaneously. No voter observes another current vote. The complete vector becomes public only after the ballot closes.

Let

$$w \text{ in } \{0,1\}^J$$

denote the full vector of weak nonproposer votes, with one meaning yes, and let **bold1** denote the all-yes vector. If voter **j** chooses yes independently with probability p_j , then

$$Q_p(w) = \prod_j p_j^{w_j} (1-p_j)^{(1-w_j)}.$$

There are three action consequences.

1. **H=Y, w=bold1**: the current agreement passes. The payoff vector is $(r_i, (x_j), y)$ in Round-1 units.

2. $H=Y, w \neq \text{bold1}$: the ballot fails, H remains active, and the complete vector reaches an active- H unanimity Round-2 continuation.
3. $H=N$: H opts out immediately and irreversibly. It receives o_{θ} now, every weak state receives zero, and no continuation is reached.

In particular, a no vote by H is not a delayed Round-2 branch.

15.1.3 The literal continuation mapping

For each failed vector following an H yes, define the full public history

```
h2Y(w) = (h1, i, s_i, H=Y, w, failed ballot), w ≠ bold1.
```

The Round-1 assessment contains a single map

```
kappa_i : public h2 -> one full element of C2, U(h2; nu(h2)).
```

At the displayed history this map returns one exact object

```
kappa_i(h2Y(w)) = (
  nu_w,
  alpha_w,
  type/player continuation payoffs c_k, wtheta,
  type-conditional outcome distributions D_wtheta,
  all internal C2 strategies and beliefs
)
```

from `pre_recognition_C2_correspondence` in the frozen `r2_unanimity_active_h_v1.json`, evaluated at that history's recorded posterior. The value c_k, w^{θ} remains in native Round-2 units. It is multiplied by β once, and only once, when this Round-1 node evaluates the failure branch. The probabilities inside D_w^{θ} are not discounted.

Different vectors with the same posterior remain different histories. The assessment may select different continuation elements at them. Conversely, two such selections may be outcome-equivalent, but that must be established from their beliefs, strategies, type/player payoffs, and outcome distributions. A raw vector by itself is not a payoff consequence.

The map is public-history measurable and type-blind: it cannot take θ as an argument. A single returned element contains both type coordinates, all player identities, the internal strategy and belief kernels, and both terminal-outcome kernels. In particular, it is invalid to obtain c_{H, w^0} from one C2 element and c_{H, w^1} (or any weak coordinate) from another. The payoff and outcome coordinates below are projections of this one full element; they are not direct selections from the scalar recognized-proposer payoff ranges reported inside the frozen C2 document.

This mapping is not a new equilibrium-selection rule. It is the ordinary completion of an assessment at all continuation histories when the frozen continuation is set-valued.

Two selected continuations count as outcome-equivalent only if their public posterior, internal strategies and beliefs, both-type all-player payoff vectors, and both-type terminal-outcome kernels agree. A common node label, passage status, or scalar payoff does not establish equivalence.

15.1.4 Beliefs and Bayes consistency

Let $\rho_i(s)$ be the weak voters' belief at the ballot after proposal s . Because a weak proposer does not observe θ , every proposal in the support of its strategy satisfies

```
rho_i(s) = mu.
```

At a globally off-path proposal, $\rho_i(s)$ is a stated assessment component.

Suppose a proposal belongs to the support of the type-independent proposal measure $\sigma_i, Q_p(w) > 0$, and

$$d = (1-\mu)h_0 + \mu h_1 > 0,$$

where h_{θ} is type θ 's H-yes action. Then a reached failed history satisfies Bayes' rule:

$$\nu_w = \mu h_1 / [(1-\mu)h_0 + \mu h_1].$$

The weak-vector probability cancels because the weak strategies are type-independent. Thus all positive-probability failed vectors after the same proposal and $H=Y$ have the same posterior, although they may retain different $\kappa_i(h_2^Y(w))$ selections because the public histories differ.

The formula has three distinct implications which must not be collapsed:

$$\begin{aligned} h=(1,1) &\rightarrow \nu_w=\mu, \\ h=(1,0) &\rightarrow \nu_w=0, \\ h=(0,1) &\rightarrow \nu_w=1, \end{aligned}$$

whenever the relevant conditioning event has positive probability. In particular, assigning $\nu_w=\mu$ after a separating $H=Y$ event violates Bayes' rule. The posterior in this calculation, the posterior field of $\kappa_i(h_2)$, its frozen-C2 membership index, and the payoff/outcome projections must all be the same number.

If the proposal, the $H=Y$ event, or the vector has zero equilibrium probability, its successor belief is off path and must be stated explicitly. Both type-contingent strategy and payoff vectors remain recorded when the public belief is zero or one.

A current weak voter evaluates a deviation under its current belief ρ . It anticipates the strategies contained in the action-specific $\kappa_i(h_2)$, and therefore uses their type-conditional payoff vectors. A recognized proposer evaluates a proposal deviation under the true preproposal belief μ , not an off-path ballot belief assigned after that deviation.

15.1.5 Payoff consequences at a fixed proposal

Conditional on type θ , define the payoff under $H=Y$ by

$$\begin{aligned} G_i^{\theta}(w) &= r_i && \text{if } w=\text{bold1}, \\ &\quad \beta c_{i,w}^{\theta} && \text{if } w \neq \text{bold1}; \\ G_j^{\theta}(w) &= x_j && \text{if } w=\text{bold1}, \\ &\quad \beta c_{j,w}^{\theta} && \text{if } w \neq \text{bold1}; \\ G_H^{\theta}(w) &= y && \text{if } w=\text{bold1}, \\ &\quad \beta c_{H,w}^{\theta} && \text{if } w \neq \text{bold1}. \end{aligned}$$

Under $H=N$, every weak payoff is zero and H receives o_{θ} . These definitions expose every payoff date. There is no β inside c , none on current passage, and none on immediate opt-out.

15.1.6 Hegemon's exact ballot incentive

15.1.6.1 Lemma 1 (full-vector H response) At a fixed proposal and weak behavioral profile, type θ 's expected payoff difference between yes and no is

$$\begin{aligned} \Delta_H(\theta) &= Q_p(\text{bold1})(y - o_{\theta}) \\ &\quad + \sum_{w \neq \text{bold1}} Q_p(w)(\beta c_{H,w}^{\theta} - o_{\theta}). \end{aligned}$$

Its unique prescribed action is

$$\begin{aligned} h_{\theta} &= 1 \quad \text{if } \Delta_H(\theta) \geq 0, \\ h_{\theta} &= 0 \quad \text{if } \Delta_H(\theta) < 0. \end{aligned}$$

Proof. Under yes, the all-yes vector implements the current payment y , and every other vector imports its own discounted continuation payoff. Under no, every vector gives the immediate payoff o_{θ} . Subtracting and averaging gives the displayed expression. The two actions always differ in inclusion, immediate opt-out, or active

continuation status. The comparison is therefore relevant with probability one; equality selects yes and excludes mixing. QED.

15.1.6.2 Corollary 1 (the direct cutoff is local to sure passage) If $Q_p(\text{bold1})=1$, Lemma 1 reduces to $h_{\theta} = 1_{\{y \geq o_{\theta}\}}$.

If any failure vector has positive probability, this reduction is generally false. For example, take $o_0=0.2$, $o_1=0.6$, $\beta=0.9$, $y=0.1$, and a sure weak failure. A valid Round-2 pooling element can pay each type of H 0.6. The low type's difference is

$$0.9(0.6)-0.2=0.34>0,$$

so it votes yes even though $y < o_0$; the high type's difference is $0.54-0.60 < 0$, so it opts out. This is a fixed-proposal ballot implication, not a claim that this proposal is optimal for the recognized proposer.

If $o_0=0$, every payoff under $H=Y$ is nonnegative. Lemma 1 then makes type zero vote yes at every proposal, with equality resolved in favor of yes.

15.1.7 Every weak voter's exact ballot incentive

Fix weak voter j . For an action a in $\{0,1\}$ and other-voter vector w_{-j} , define its payoff conditional on $H=Y$ by

$$g_j^{\theta}(a, w_{-j}) = \begin{cases} x_j & \text{if } (a, w_{-j}) = \text{bold1}, \\ \beta c_j, (a, w_{-j})^{\theta} & \text{otherwise.} \end{cases}$$

Let $\lambda_0(\rho)=1-\rho$ and $\lambda_1(\rho)=\rho$. Voter j 's expected payoff difference is

$$\Delta_j = \sum_{\theta} \lambda_{\theta} \lambda_{\theta}(\rho) h_{\theta} \sum_{w_{-j}} Q_{p,-j}(w_{-j}) [g_j^{\theta}(1, w_{-j}) - g_j^{\theta}(0, w_{-j})].$$

The terms with $H=N$ are zero because both weak actions end at the same immediate opt-out outcome.

Let R_j be the probability, under the same belief and other-player strategies, that $H=Y$ and the two action-specific outcome signatures differ. The signature includes current implementation or the selected continuation's beliefs, strategies, type/player payoffs, and outcome distributions.

15.1.7.1 Lemma 2 (full-vector weak response) The exact response is

$$\begin{aligned} \text{if } R_j > 0 \text{ and } \Delta_j \geq 0: & p_j = 1; \\ \text{if } R_j > 0 \text{ and } \Delta_j < 0: & p_j = 0; \\ \text{if } R_j = 0: & p_j \text{ may be any number in } [0,1]. \end{aligned}$$

Proof. The displayed Δ_j enumerates both types and every vector of the other simultaneous votes. If relevance has positive probability, the frozen equality convention selects yes at a tie; otherwise the strict payoff sign selects the corresponding action. If relevance is zero, the two induced signatures, including payoffs, agree on the assessment support, so the expected difference is zero and either action is a completion. QED.

Two important consequences follow.

1. When all other weak voters choose yes and $H=Y$, voter j compares current x_j with its discounted continuation payoff after the single revealed no.
2. When another weak no already guarantees current failure, j may still be relevant: its two votes reveal different full histories and can induce two different $\kappa_{p,i}(h_2)$ selections. Hence an unchanged quota outcome does not remove the continuation term.

For example, if both actions fail but their selected native continuation payoffs to j are 0.4 after yes and 0.1 after no, the contribution is $\beta(0.4-0.1) > 0$. Reversing the two selections reverses the sign. This is why the Round-1 ballot cannot inherit the terminal Round-2 weak response formula.

15.1.8 Exact fixed-proposal characterization

For every feasible proposal, define a complete ballot element

```
A_i(s) = (
  rho_i(s),
  h_0, h_1,
  (p_j)_{j in J_i},
  {(nu_w, kappa_i(h2^Y(w))): w != bold1}
).
```

15.1.8.1 Proposition 1 (necessary and sufficient fixed-proposal fixed point) $A_i(s)$ is sequentially rational at every current ballot information set and every reached continuation if and only if all four conditions hold:

1. each $\kappa_i(h2^Y(w))$ is a literal member of the frozen $C2, U(h2^Y(w); \nu_w)$ correspondence at its recorded posterior;
2. every positive-probability failed history satisfies the Bayes formula in Section 3;
3. both type actions satisfy Lemma 1; and
4. every weak behavioral probability satisfies Lemma 2 over the complete vector space.

Proof. Necessity follows directly from continuation membership, Bayes consistency, and one-shot deviations at each simultaneous ballot information set. For sufficiency, the selected $\kappa_i(h2)$ is already a complete PBE of every reached Round-2 continuation. Lemmas 1 and 2 eliminate every current ballot deviation, and the belief conditions supply consistency wherever Bayes applies. There are no other actions between the current proposal and the terminal outcome or imported continuation. QED.

This proposition is a closed assessment-level characterization even though it is not a scalar formula. At fixed N and proposal there are $2^{(N-2)}-1$ failed-vector slots, two current H actions, and $N-2$ independent current weak behavioral probabilities. Each slot nevertheless contains a complete frozen $C2$ assessment element, not a scalar or a finite-dimensional replacement for it.

15.1.8.2 Population boundary When $N=3$, there is one weak nonproposer. A single no changes current passage to continuation, so its action is generally relevant. When $N \geq 4$, two sure no votes make current passage impossible after either voter's unilateral switch. Nevertheless, their actions have zero relevance only if the two action-specific continuation elements are outcome-equivalent. The population count alone does not prove that equivalence.

15.1.9 Proposal optimality and the full proposer assessment

A recognized-proposer assessment contains:

1. a Borel probability measure σ_i on the feasible proposal set;
2. one complete $A_i(s)$ satisfying Proposition 1 at every feasible proposal, including every deviation; and
3. the resulting full type/player payoff and outcome kernels.

The maps $\rho_i(s)$, $h_i, \theta(s)$, $p_{i,j}(s)$, and all current payoff and outcome kernels are Borel measurable. The continuation selector $\kappa_i(h2)$ is measurable in the public history and type-blind. These are parts of one assessment; payoff coordinates, posterior coordinates, and outcomes cannot be assembled from different proposal-contingent selections.

For each proposal-assessment pair, define the type-conditional payoffs

```
U_i^theta
= sum_w Q_p(w) h_theta
  [1{w=bold1}r_i + 1{w!=bold1}beta c_i, w^theta],

U_j^theta
= sum_w Q_p(w) h_theta
  [1{w=bold1}x_j + 1{w!=bold1}beta c_j, w^theta],
```

$$\begin{aligned}
U_H^\theta &= (1-h_\theta)o_\theta \\
&+ h_\theta[\\
&\quad Q_p(\text{bold1})y \\
&\quad + \sum_{w \neq \text{bold1}} Q_p(w)\beta c_{H,w}^\theta \\
&].
\end{aligned}$$

The recognized proposer's objective and the subsequent tie-break quantity are

$$\begin{aligned}
V_i(s, A; \mu) &= (1-\mu)U_i^0 + \mu U_i^1, \\
\bar{H}(s, A; \mu) &= (1-\mu)U_H^0 + \mu U_H^1.
\end{aligned}$$

Every support proposal must maximize V_i against every feasible proposal under that proposal's preassigned assessment element. Among all proposer maximizers, support proposals must minimize $\bar{H}$. If both quantities tie, different weak payment vectors, continuation selections, beliefs, and outcome distributions remain in the correspondence. Proposal mixing is possible only among such double ties.

15.1.9.1 Proposition 2 (necessary and sufficient recognized-proposer PBE) A complete proposer assessment is a PBE if and only if:

1. every proposal-indexed ballot element satisfies Proposition 1;
2. proposal and successor beliefs satisfy Section 3; and
3. the proposal support satisfies maximization and the stated tie-break.

Proof. Proposition 1 closes all ballot and Round-2 deviations. Condition 3 is exactly sequential rationality at the remaining Round-1 proposal node. These conditions are therefore sufficient. Any PBE must specify behavior and beliefs after every proposal, must be sequentially rational at every such ballot, and must make the recognized proposer's support optimal, so they are necessary. QED.

The proposition also gives the exact existence test: the PBE correspondence is nonempty precisely when this proposal-indexed fixed point has a nonempty maximizer set. Sections 9.4 and 9.5 give attained constructions for $N \geq 4$ and $N=3$, respectively. The $N=3$ construction handles the frozen C2 nonclosed boundary without adding its missing zero endpoint.

15.1.10 Constructive and diagnostic subclasses

These subclasses are consequences of Propositions 1 and 2. None replaces the full assessment.

15.1.10.1 Secured current passage for both types Suppose every weak nonproposer votes yes and $y \geq o_1$. Lemma 1 then gives $h_0 = h_1 = 1$. Voter j 's current yes implements x_j ; its no reaches the single-no continuation. Its condition is

$$\begin{aligned}
&\text{sum_theta } \lambda_{\theta}(\rho) \\
&[x_j - \beta c_{j,w}(j=0, \text{others}=1)^\theta] \geq 0.
\end{aligned}$$

If every voter condition holds, the ballot fixed point implements a current agreement for both types. It is an on-path proposer PBE only if the proposal also satisfies global proposal optimality and the tie-break.

15.1.10.2 Secured current passage for the low type only Suppose $\rho < 1$, all weak nonproposers vote yes, and

$$o_0 \leq y < o_1.$$

Then $h_0 = 1, h_1 = 0$. Each weak condition reduces to

$$\begin{aligned}
&(1-\rho) \\
&[x_j - \beta c_{j,w}(j=0, \text{others}=1)^0] \geq 0.
\end{aligned}$$

The low type is included and the high type opts out immediately. Again, proposal optimality is a separate necessary condition.

15.1.10.3 Flat-continuation failure ballot Let $N \geq 4$ and let at least two weak voters choose no with probability one. Suppose the assessment assigns outcome-equivalent C2 elements across every failed history connected by one weak voter's switch on the support of the other votes. Every such weak action then has zero relevance. Passage has probability zero, and Lemma 1 becomes

$h_{\theta}=1$
iff $\beta \sum_w Q_p(w) c_{H,w}^{\theta} \geq o_{\theta}$.

This supplies a constructive ballot completion that may retain H for continuation even at a current payment below its outside option. It becomes a proposer-stage PBE only if its proposer value defeats every proposal deviation and then satisfies the tie-break.

15.1.10.4 Proposition 3 (universal existence and attainment for $N \geq 4$) For every $N \geq 4$, every μ in $[0,1]$, every β in $(0,1]$, and every admissible pair of outside options, the Round-1 unanimity PBE correspondence is nonempty.

Proof. At every proposal set every weak nonproposer to no. Because there are at least two such voters, a unilateral switch by any one of them still fails. Set

$h_{\theta} = 1\{\beta o_{\theta} \geq o_{\theta}\}$.

The continuation selector is constructed only after its public posterior is fixed. At every positive-probability failed history after $H=Y$, compute

$\nu(h_2) = \mu h_1 / [(1-\mu)h_0 + \mu h_1]$

and select from the frozen pre-recognition $C2, U(h_2; \nu(h_2))$ correspondence the canonical coordinated-failure element: after every R2 proposal use the deterministic all-no weak completion, let both R2 H types choose yes, and choose a fixed feasible R2 proposal at each recognition node. This explicit element exists for $N \geq 4$, is measurable in its sole varying belief coordinate ν , pays every weak state zero, and records both H coordinates $c_{H,w}^{\theta} = o_{\theta}$. At globally off-path failed histories, state a posterior and choose the same class from the C2 correspondence at that stated posterior. Across histories connected by one weak switch after a common proposal, choose elements with the same posterior and outcome-equivalent strategy, belief, both-type payoff, and terminal-outcome kernels. This defines one public-history-measurable, type-blind κ_i , not one selection per private type.

The Bayes cases are explicit. If $\beta=1$, then $h=(1,1)$ and every reached failure has posterior μ . If $\beta<1$ and $o_0=0$, then $h=(1,0)$ and every reached $H=Y$ failure has posterior zero—not μ . If $\beta<1$ and both outside options are positive, $H=Y$ failures are off path.

Passage is impossible. Lemma 1 therefore gives

$h_{\theta}=1$ iff $\beta o_{\theta} \geq o_{\theta}$.

Thus H votes yes only when $\beta=1$ or $o_{\theta}=0$; in either case its payoff is exactly o_{θ} . Otherwise it chooses immediate opt-out and again gets o_{θ} . Every feasible proposal consequently gives the proposer zero and gives H its expected outside payoff. Assign this same completion after every proposal. All proposals tie in the proposer objective, and they also tie under the frozen H-payoff rule. Any feasible support proposal therefore satisfies Proposition 2. The construction is an actual assessment element, not a limit, so nonemptiness is attained. QED.

15.1.10.5 Proposition 4 (universal existence and attainment for $N=3$) For $N=3$, every admissible μ , β , o_0 , and o_1 admits an attained Round-1 unanimity PBE.

Proof. There is one weak nonproposer. Write

$G=1-o_1$, $D=1-o_0$.

We first define three literal frozen C2 elements used below.

1. If $G>0$, let $c_G=G/2$. For every posterior ν , let κ^G_{ν} be the symmetric pre-recognition C2 element in which each R2 recognized proposer offers $y_2=o_1$, pays zero to the other weak state, and retains G . The

frozen $N=3$ C2 theorem makes this an attained pooling PBE for every μ . Each weak identity has both type-conditional pre-recognition payoff c_G , while both H coordinates equal o_1 .

2. If $G=0$, let $c_0=D/4$. At posterior zero, let κ^0_0 symmetrically assign each R2 recognized proposer the separating proposal

$$y_2=o_0+D/2, \quad x_2=0, \quad r_2=D/2.$$

This is an attained member of the frozen interval $(0, D]$. Each weak identity's low-type pre-recognition payoff is c_0 , and H's low coordinate is $o_0+D/2$.

3. Still when $G=0$, let κ^0_1 be the posterior-one C2 element with $y_2=1$ and all weak payments zero. It gives both weak identities zero in both type coordinates and both H coordinates one.

All three are complete public-history-indexed, type-blind elements containing both type coordinates. We now cover the primitive domain.

Case 1: $G>0$ and $\beta<1$. On path propose

$$(y, x) = (o_1, \beta c_G).$$

Both H types and the weak voter choose yes. A weak no has posterior μ and selects κ^G_μ , so the voter is paid βc_G under both actions and yes is selected at equality. The proposer obtains

$$V = G - \beta c_G = G(1 - \beta/2) > 0.$$

After every off-path proposal set $\rho=1$, choose weak no, and select κ^G_1 after $H=Y$. High H strictly opts out because $\beta o_1 < o_1$; hence the weak no has zero relevance. Low H may choose either best response, but a deviation pays the proposer at most

$$(1 - \mu)\beta c_G \leq \beta G/2 < G(1 - \beta/2) = V.$$

Thus the on-path proposal is a strict optimum.

Case 2: $G>0$ and $\beta=1$. Define

$$P = c_G, \quad R = D - c_G, \quad S = (1 - \mu)R.$$

Use pooling $(y, x) = (o_1, c_G)$ when $P > S$; use separation $(y, x) = (o_0, c_G)$ when $S \geq P$. The weak voter is indifferent between its current payment and κ^G_μ under pooling or κ^G_0 after the low-type separating yes, and therefore chooses yes.

At every off-path proposal set $\rho=1$ and select κ^G_1 after a weak no. If $x < c_G$, assign weak no. Both H types then choose yes into the continuation, and the deviation gives the proposer exactly $c_G = P$. If $x \geq c_G$, assign weak yes. A pooling deviation has residual at most $1 - o_1 - c_G = P$; a separating deviation has expected residual at most $(1 - \mu)(1 - o_0 - c_G) = S$; all other deviations yield zero. Thus no deviation exceeds $\max\{P, S\}$. When $P = S$, separation pays H $(1 - \mu)o_0 + \mu o_1 < o_1$ whenever $\mu < 1$, so the authorized proposer tie-break selects it. At $\mu = 1$, $S = 0 < P$. This closes the equality boundary.

Case 3: $G=0$, $\mu < 1$, and $\beta < 1$. On path propose

$$(y, x) = (o_0, \beta c_0).$$

Low H and the weak voter choose yes; high H opts out. A weak no following low H yes has Bayes posterior zero and selects κ^0_0 , so the voter is indifferent and yes is selected. The proposer obtains

$$V = (1 - \mu)(D - \beta c_0) > 0.$$

After every off-path proposal set $\rho=1$, assign weak no, and select κ^0_1 . High H strictly opts out because $\beta < 1$; the weak vote has zero relevance. Any low-type continuation gives the proposer zero under κ^0_1 , so every deviation pays zero.

Case 4: $G=0$, $\mu < 1$, and $\beta=1$. On path propose $(y, x) = (o_0, c_0)$, use κ^0_0 after a weak no, and obtain

$$V = (1 - \mu)(D - c_0) > 0.$$

At every off-path proposal with $y < 1$, set $\rho = 1$, select κ^0_1 , and assign a measurable probability $p(s) > 0$ satisfying, whenever the relevant denominator is nonzero,

$$\begin{aligned} p(s) &\leq 1/2, \\ p(s) &\leq D/[2(1-y)], \\ p(s) &\leq (D-c_0)/[2r(s)]. \end{aligned}$$

The first two inequalities make low H accept and high H reject; because the weak ballot belief assigns probability one to high H, its relevance is zero and this positive $p(s)$ is a valid completion. The last inequality bounds the true-prior deviation payoff by $V/2$. At $y=1$, feasibility forces zero weak residual. Hence the on-path proposal is a strict optimum. The displayed minimum of continuous positive functions (with a term omitted when its denominator is zero) is Borel measurable.

Case 5: $G=0$ and $\mu=1$. Propose $(y,x)=(1,0)$ and assign all yes. It passes with zero weak payoff and gives H one. If $\beta < 1$, complete every off-path proposal with $\rho = 1$, weak no, and κ^0_1 ; high H then opts out. If $\beta = 1$ and $y < 1$, choose any strictly positive weak-yes probability, so high H opts out and the weak action has zero relevance; at $y=1$, choose weak yes, and feasibility forces zero proposer residual. Therefore every proposer payoff is zero and every supported-type H payoff is one. The proposal and H-payoff tie conventions are satisfied for every β .

The cases are exhaustive. Crucially, when $G=0$ the construction never uses the unattained zero endpoint of the posterior-below-one C2 correspondence: it uses the positive attained κ^0_0 at every reached separating weak-no history, and uses κ^0_1 only after globally off-path proposals. QED.

15.1.11 Gifts, mixing, and multiplicity

No coalition size is imposed. A positive named payment is feasible even when its recipient is not needed to change passage. Deleting that payment creates a new proposal information set with its own belief, ballot fixed point, and continuation mapping; it cannot be assumed to inherit the original proposal's completion. Hence no zero-gift conclusion follows from proposal optimality without an additional proof.

The exact mixing possibilities are:

- H never mixes because its comparison is always relevant and equality selects yes;
- a weak voter may mix only when $R_j = 0$;
- imported C2 elements retain all internal mixing permitted by the frozen interface; and
- the recognized proposer may mix only across elements tying in both V_i and H_{bar} .

Identity symmetry is not imposed. Isomorphic recognized-proposer nodes may select different PBE elements.

15.1.12 Validated pre-recognition Round-1 interface ($N \geq 3$)

A complete Round-1 assessment α selects one recognized-proposer PBE element

$$e_i = (\sigma_i, \rho_i, h_i, p_i, \kappa_i)$$

from Proposition 2 for every $i=1, \dots, m$. Selections may be asymmetric by identity. For every player identity k and type coordinate θ , first integrate over the proposal measure actually used by recognized proposer i :

$$\begin{aligned} E_{\sigma_i}[U_k^i(\theta; e_i)] \\ = \int U_k^i(\theta; s, e_i) \sigma_i(ds). \end{aligned}$$

Only then integrate recognition. The full type-by-identity interface is

$$\begin{aligned} C_1, U^\alpha(h1)_{\{W_k\}}(\theta) \\ = (1/m) \sum_i E_{\sigma_i}[U_{\{W_k\}}^i(\theta; e_i)] \end{aligned}$$

for weak state k , and

$$\begin{aligned} C_1, U^\alpha(h1)_H(\theta) \\ = (1/m) \sum_i E_{\sigma_i}[U_H^i(\theta; e_i)]. \end{aligned}$$

The ex-ante weak payoff is a separate final projection:

$$\begin{aligned}
V_{\{W_k, U\}^\alpha(h1; \mu)} \\
&= (1-\mu) C_{1, U^\alpha(h1)_{\{W_k\}}(0)} \\
&\quad + \mu C_{1, U^\alpha(h1)_{\{W_k\}}(1)}.
\end{aligned}$$

It is not folded into the C1 type coordinates. A cross-weak average may be reported descriptively, but it does not replace the vector $(V_{\{W_k, U\}^\alpha})_k$ for collective entry. For each **theta**, the same **sigma_i**, recognition weights, and **alpha** are applied to current passage, immediate opt-out, payments, beliefs, and every type-conditional terminal distribution imported through **kappa_i**. Equal values of **mu** at different **h1** do not force a common selection.

This repaired export covers $N \geq 3$, including the attained $N=3$ construction in Proposition 4. Its exact hash was accepted by two independent rereviews and is consumed downstream only through the frozen R1 batch.

15.1.13 Selection-free bounds

Every weak payoff is nonnegative. Under current passage, total weak payoff is $1-y \leq 1$; under continuation, the frozen type-conditional total is at most one and $\beta \leq 1$; under H no it is zero. Therefore, type by type and ex ante,

$$\begin{aligned}
0 &\leq U_k \leq 1, \\
0 &\leq \sum_k U_k \leq 1.
\end{aligned}$$

At every reached ballot, type **theta** of H can choose immediate opt-out. Sequential rationality and the unit upper bound therefore imply

$$o_{\text{theta}} \leq C_{1, U, H(\text{theta})} \leq 1.$$

These bounds do not select an assessment and cannot replace the payoff correspondence.

15.1.13.1 The $N=3$ nonclosed endpoint is preserved, not filled When $o_1=1$ and $\nu < 1$, the frozen $N=3$ C2 proposer-value set remains $(0, L(\nu)]$: zero is an unattained infimum. Proposition 4 does not optimize over its closure. It chooses the interior attained value $D/2$ at posterior zero, which yields $c_0=D/4$ to each weak identity before recognition. At posterior one, the separate frozen $y_2=1$ element is itself attained. Thus the R1 existence proof closes while the C2 endpoint remains nonclosed.

15.1.14 Survival findings for historical outcome motifs

The following statements are diagnostic only.

- If the historical label **P** denotes a both-type current agreement, its outcome motif survives conditionally through Section 9.1. No historical **P** formula is imported.
- If the historical label **L** denotes low-type current passage followed by high-type opt-out, its outcome motif survives conditionally through Section 9.2. No historical **L** formula is imported.
- If the historical label **R** denotes deliberate failure followed by an active-H continuation, its outcome motif survives conditionally only when the full failed-vector mapping satisfies Propositions 1 and 2, as in Section 9.3. Immediate H opt-out is not this continuation motif.
- A theorem reducing every current PBE to three scalar labels does not survive. The exact object also retains immediate opt-out histories, vector-specific C2 selections, gifts, zero-relevance completions, double-tied proposal mixtures, beliefs, and full outcome distributions. Enlarging three labels until they contain all these objects would merely rename, not reduce, the assessment-level correspondence.

No historical formula or rejected-history argument is used to obtain these findings.

15.1.15 Claims ledger

- **Complete action-specific C2 import.**
 - **Method:** direct frozen-interface construction
 - **Status:** proved
- **Full-vector H incentive.**
 - **Method:** enumeration of every weak vector
 - **Status:** proved

- **Full-vector weak incentive.**
 - **Method:** enumeration of both types and every other-vote vector
 - **Status:** proved
- **Fixed-proposal fixed-point characterization.**
 - **Method:** necessity and sufficiency proof
 - **Status:** proved
- **Proposal-stage PBE characterization.**
 - **Method:** one-shot proposal deviation proof
 - **Status:** proved
- **Bayes update and off-path classification.**
 - **Method:** direct probability calculation
 - **Status:** proved
- **Exactly-one discount and immediate opt-out dating.**
 - **Method:** branchwise payoff ledger
 - **Status:** proved
- **Pre-recognition E_{σ} , recognition integration, and separate μ projection for $N \geq 3$.**
 - **Method:** law of iterated expectations
 - **Status:** proved
- **Selection-free weak and H bounds.**
 - **Method:** terminal accounting and H deviation option
 - **Status:** proved
- **Conditional survival of three outcome motifs.**
 - **Method:** constructive subclasses
 - **Status:** proved
- **Global three-label reduction.**
 - **Method:** counter-scope from exact correspondence
 - **Status:** rejected
- **Universal existence and attainment for $N \geq 4$.**
 - **Method:** Bayes-aligned coordinated-failure construction at every proposal
 - **Status:** proved
- **Universal existence and attainment for $N = 3$.**
 - **Method:** exhaustive attained construction preserving the nonclosed $C2$ endpoint
 - **Status:** proved
- **Validated downstream export domain.**
 - **Method:** explicit machine gate
 - **Status:** $N \geq 3$
- **Full-vector enumeration, Bayes/membership negatives, map/measurability export, and mutation guards.**
 - **Method:** node-specific R verifier, 37/37 PASS
 - **Status:** checked numerically
- **Independent formal/adversarial acceptance.**
 - **Method:** two exact-hash read-only reviews
 - **Status:** proved by review bundle
- **Inclusion in standalone output.**
 - **Method:** post-freeze render
 - **Status:** checked separately

15.1.16 Invalidation

Any byte change to the Gate 0 bundle, the frozen Round-2 batch, or the $C2$ -U interface invalidates this frozen interface. The same is true of any change to the simultaneous information structure, full-vector publication, proposal budget, immediate opt-out, payoff dates, Bayes discipline, or either frozen tie convention. Every later entry or comparison node consuming this interface must then be reopened.

15.2 Round 1 majority with an active hegemon

15.2.1 Scope and frozen continuations

This note derives only `r1_majority`. It consumes, without repairing or selecting inside the proof, the independently reviewed Round-2 batch

R2 batch	sha256:00f84d42...
C2 majority, active H	sha256:a4ae73e9...
C2 majority, weak only	sha256:e0ec6cd3...

The solution concept is PBE under the Gate 0 response-at-equality rule. A ballot is simultaneous and sealed. No coalition restriction, zero-gift restriction, common completion, or scalar continuation is imposed. Every C2 payoff below is in native Round-2 units and is multiplied by `beta` exactly once only on a branch that reaches Round 2.

15.2.2 State, proposal, beliefs, and assessment objects

There are $N=m+1 \geq 3$ states, H and m weak states. Before recognition the public state is

`(M, round=1, H=active, full h1, mu(h1)).`

A weak proposer i is recognized uniformly. After recognition it proposes

$s=(y, (x_j)_{\{j \neq i\}}),$	$0 \leq y \leq y_{\text{bar}},$
$x_j \geq 0,$	$X = \sum_{\{j \neq i\}} x_j \leq 1 - y,$
$r_i = 1 - y - X.$	

There are $n=m-1=N-2$ weak nonproposers. The proposer is counted as yes. A complete nonproposer vector is $\mathbf{a}$ in $\{0,1\}^n$, with

`z(a)=1+sum_j a_j`

total weak yes votes. The majority quota is $q=\text{floor}(N/2)+1$.

The ballot belief after proposal $\mathbf{s}$ is $\text{rho}(\mathbf{s})$. On the support of the uninformed weak proposer's strategy, $\text{rho}(\mathbf{s})=\mu$. At a globally off-path proposal, $\text{rho}(\mathbf{s})$ is an explicit assessment component. Voters use $\text{rho}(\mathbf{s})$; the proposer evaluates every deviation under the true preproposal μ . These two objects are not interchangeable.

Let $p_j(\mathbf{s})$ be weak voter j 's independent probability of yes. The response rule below makes each H type's action pure; write it as $\mathbf{h}_{\text{theta}}(\mathbf{s})$ in $\{Y,N\}$. For every complete action-specific failed history, the assessment also fixes

`kappa_A(h2) in C2_M_active(h2),`
`kappa_0(h2) in C2_M_W0(h2).`

The first selection contains the complete type-by-identity payoff vector, proposal and ballot assessment, terminal-signature distribution, and beliefs from the frozen active-H interface. The second contains the complete weak-only object from the frozen post-opt-out interface. Selections are made as parts of the strategy assessment before any current ballot is observed; they are not choices granted to a voter after seeing the vector. Equal raw successor labels do not identify selections at distinct full histories.

Both selection maps are public-measurable and type-blind. Their argument is the complete public history $\mathbf{h2}$, never the privately known theta . If one public history is compatible with both types, one and the same selected C2 element governs both types and carries its own type-conditional payoff and outcome coordinates. Thus notation such as $C^A_{\{k, \text{theta}\}}(\mathbf{h2})$ selects the theta coordinate of $\text{kappa}_A(\mathbf{h2})$; it does not denote a type-contingent choice of kappa_A . The same restriction applies to kappa_0 .

At a positive-probability history following H action d , Bayes' rule gives

`nu(h2)=rho Pr(d|theta=1,s) /`
`[(1-rho)Pr(d|theta=0,s)+rho Pr(d|theta=1,s)].`

The weak vector has no type likelihood conditional on $\mathbf{s}$. A history that has zero probability under the complete local strategy receives an explicit off-path posterior. This includes a history reached only after a pure-action

deviation. Each $\kappa_A(h_2)$ must belong to the active-H C2 correspondence at that exact full history and posterior. $\kappa_0(h_2)$ remains indexed by the full history even though type is payoff-irrelevant after opt-out.

15.2.3 Exhaustive transition and mixed-date payoff map

For player k , type θ , H action d , and weak vector a , define $U_k^d(\theta, a; s)$ in Round-1 units as follows.

- Y.
 - **Weak count:** $z \geq q-1$
 - **Gate 0 branch:** PR04, current passage with H
 - **Round-1 payoff:** H: y ; proposer r_i ; named weak j : x_j
- Y.
 - **Weak count:** $z \leq q-2$
 - **Gate 0 branch:** PR05, active-H C2
 - **Round-1 payoff:** $\beta C^A_{\{k, \theta\}}(h_2^Y a)$
- N.
 - **Weak count:** $z \geq q$
 - **Gate 0 branch:** PR06, current weak-only passage
 - **Round-1 payoff:** H: o_θ ; proposer $1-X$; named weak j : x_j
- N.
 - **Weak count:** $z \leq q-1$
 - **Gate 0 branch:** PR07, weak-only C2
 - **Round-1 payoff:** H: o_θ ; weak k : $\beta C^0_k(h_2^N a)$

Thus an H no pays o_θ immediately and exactly once. It never receives βC^M_W and never receives $o_\theta + \beta o_\theta$. The weak players, and only the weak players, receive βC^M_W on PR07. On PR05 every player receives the active-H continuation discounted once. Current passage uses current payoffs and no discount.

For a fixed proposal assessment, let

$$P_p(a) = \prod_j p_j^{a_j} (1-p_j)^{(1-a_j)}.$$

The type-conditional payoff and full outcome distribution are obtained by integrating the displayed four-branch vector over a ; the true proposer payoff is

$$G_i(s; \mu) = (1-\mu) E_p[U_i^{\{h_0\}}(0, a; s)] + \mu E_p[U_i^{\{h_1\}}(1, a; s)]. \quad (1)$$

The corresponding G_H and every identity-specific G_j use the same joint distribution. The interface retains those vectors and the PR04–PR07, inclusion, opt-out, implemented-payment, public-history, and C2-signature coordinates; equation (1) is only a scalar projection.

15.2.4 H's exact voting condition

Fix s , p , all action-specific beliefs, and both C2 selection maps. For type θ , H's monetary difference between yes and no is

$$\begin{aligned} \Delta_H(\theta) &= \sum_{\{a: z(a) \geq q-1\}} P_p(a) [y - o_\theta] \\ &+ \sum_{\{a: z(a) \leq q-2\}} P_p(a) [\beta C^A_{\{H, \theta\}}(h_2^Y a) - o_\theta]. \end{aligned} \quad (H-IC)$$

κ_0 does not enter H-IC: no realizes o_θ now irrespective of whether the weak ballot currently passes. H's action changes at least opt-out status for every a , including profiles where the weak quota is already secured and profiles where both actions have the same monetary payoff. H therefore has relevance probability one and the prescribed action is

$$h_\theta = Y \text{ iff } \Delta_H(\theta) \geq 0; \text{ otherwise } h_\theta = N. \quad (H-BR)$$

There is no mixed H action at equality. Unlike terminal R2, this is generally not a global threshold in y : every failed-vector continuation selection in (H-IC) can matter.

15.2.5 Each weak voter's exact voting condition

Fix weak nonproposer j and a vector $\mathbf{a}_{-j}$; let

$$z_0 = 1 + \sum_{k \neq j} \mathbf{a}_k$$

be the total weak yes count when j votes no. Define the action-specific monetary differences $\delta_j^d(\theta, \mathbf{a}_{-j})$.

For H yes,

$$\begin{aligned} \delta_{j,Y} &= 0, & \text{if } z_0 \geq q-1; \\ \delta_{j,Y} &= x_j - \beta C^A_{j,\theta}(\mathbf{h}_Y^2_{\mathbf{a}_{-j}=N}), & \text{if } z_0 = q-2; \\ \delta_{j,Y} &= \beta [C^A_{j,\theta}(\mathbf{h}_Y^2_{\mathbf{a}_{-j}=Y}) - C^A_{j,\theta}(\mathbf{h}_Y^2_{\mathbf{a}_{-j}=N})], & \text{if } z_0 \leq q-3. \end{aligned} \quad (\text{W-Y})$$

For H no,

$$\begin{aligned} \delta_{j,N} &= 0, & \text{if } z_0 \geq q; \\ \delta_{j,N} &= x_j - \beta C^0_j(\mathbf{h}_N^2_{\mathbf{a}_{-j}=N}), & \text{if } z_0 = q-1; \\ \delta_{j,N} &= \beta [C^0_j(\mathbf{h}_N^2_{\mathbf{a}_{-j}=Y}) - C^0_j(\mathbf{h}_N^2_{\mathbf{a}_{-j}=N})], & \text{if } z_0 \leq q-2. \end{aligned} \quad (\text{W-N})$$

Empty ranges are ignored. In particular, the structural H-yes toggle is empty at $N=3$. Let $P_{-j}(\mathbf{a}_{-j})$ be the independent probability of the other weak vector. The weak voter's exact expected monetary difference is

$$\begin{aligned} D_j &= \sum_{\theta} \rho_{\theta} \sum_{\mathbf{a}_{-j}} P_{-j}(\mathbf{a}_{-j}) \\ &\quad [1_{\{\mathbf{h}_{\theta}=Y\}} \delta_{j,Y}(\theta, \mathbf{a}_{-j}) + 1_{\{\mathbf{h}_{\theta}=N\}} \delta_{j,N}(\theta, \mathbf{a}_{-j})]. \end{aligned} \quad (\text{W-IC})$$

Separately, let R_j be the probability, under the same distribution, that the two action-specific **full Gate 0 outcome signatures** differ. On a double-failure profile, this comparison is between $\kappa(\mathbf{h}_Y^2_{\mathbf{a}_{-j}=Y})$ and $\kappa(\mathbf{h}_Y^2_{\mathbf{a}_{-j}=N})$, not between two raw node labels. It includes continuation outcomes, beliefs when relevant, inclusion, opt-out, payments, and payoff distributions. The response condition is

$$\begin{aligned} \text{if } R_j > 0: & \quad p_j = 1 \text{ when } D_j \geq 0 \text{ and } p_j = 0 \text{ when } D_j < 0; \\ \text{if } R_j = 0: & \quad \text{any } p_j \text{ in } [0,1] \text{ is an explicit completion.} \end{aligned} \quad (\text{W-BR})$$

Thus an interior weak randomization can only be a zero-relevance completion. The condition does not imply zero gifts, minimal support, or common completions across proposals.

15.2.5.1 Proposition 1 (fixed-proposal ballot fixed point) For fixed $\mathbf{s}$, ρ , posterior map, and admissible action-specific C2 selection maps, a ballot profile is sequentially rational if and only if (H-BR) holds for both types and (W-BR) holds for every weak nonproposer.

Proof. The four rows in Section 2 exhaust Gate 0 transitions. Holding others' simultaneous actions fixed, subtracting H's no payoff gives (H-IC). For a weak voter, direct subtraction in the three exhaustive count regions for each H action gives (W-Y) and (W-N). Integrating at that information set gives (W-IC). Gate 0 then prescribes (H-BR) and (W-BR), including equality and zero-relevance completions. No later choice remains inside the R1 ballot. QED.

15.2.6 Exact proposal-stage PBE characterization

For recognized proposer i , a majority-R1 assessment consists of:

1. a proposal distribution σ_i ;
2. a belief map $\rho_i(s)$, equal to μ on support and explicit elsewhere;
3. type actions, independent weak ballot probabilities, and full-profile completions satisfying Proposition 1 at every proposal;
4. a posterior map at every complete public h_2 , Bayesian whenever that history has positive probability;
5. one literal $\kappa_A(h_2)$ or $\kappa_0(h_2)$ from the appropriate frozen C2 interface at every action-specific failure history;
6. the complete type-by-identity payoff and outcome distributions; and
7. proposer support contained in

$$\operatorname{argmax}_s G_i(s; \mu), \quad (\text{P-OPT})$$

with a nonempty maximum.

After proposer-payoff maximization, only maximizers minimizing $G_H(s; \mu)$ are retained. If proposer and H payoffs tie while allocations, identities, type outcomes, histories, completion maps, or C2 signatures differ, the correspondence remains. Proposal mixing is possible only among elements with the same maximal proposer payoff and the same tie-minimal H payoff.

For a possibly mixed proposal strategy, the payoff of player k , conditional on type θ , is the expectation over the proposal draw,

$$U_k^{i, \theta}(\alpha_i) = \int U_k^{\theta}(s, A_i(s)) d\sigma_i(s). \quad (\text{SIGMA})$$

Consequently, proposer maximization and the H tie-break apply to each support element, while the exported type-by-identity vector and outcome distribution integrate those double-tied elements using σ_i . Mixing never turns two nonmaximizing pure proposals into a maximizing lottery.

15.2.6.1 Proposition 2 (necessity and sufficiency) The seven conditions above are necessary and sufficient for a recognized- proposer PBE assessment, provided (P-OPT) has a maximum. They are the exact global characterization; no generally valid scalar continuation or global offer threshold exists.

Proof. Proposition 1 is necessary and sufficient at every ballot information set. Frozen C2 membership is necessary and sufficient after every failed history. Bayes' rule supplies on-path consistency. Conditional on the complete continuation strategy map, (1) is the proposer's payoff from every feasible deviation, so (P-OPT) plus the primitive tie-break is exactly its sequential-rationality condition. These conditions also construct a complete strategy and belief assessment. QED.

The explicit **argmax** clause is material: arbitrary off-path belief and completion maps can make a continuum-action payoff function discontinuous. The game supplies no compactness/continuity selection rule that would permit the proof to replace nonattainment by a supremum. The constructive subclasses below establish nonempty PBEs without such an assumption.

15.2.7 Exact closed boundary $N=3$

When $N=3$, $q=2$ and there is one weak nonproposer. H yes always implements the current agreement, so active- H C2 is never reached. H follows the direct threshold $h_{\theta}=Y$ iff $y \geq o_{\theta}$. If H no, the weak voter compares current x_j with the unique weak-only continuation $b=\beta/2$: it votes yes iff $x_j \geq b$ whenever rejection has positive ballot-belief probability; if all believed types accept, its action is a completion.

Define

$$\begin{aligned} b &= \beta/2, \\ A &= 1-b, \\ C &= (1-\mu)(1-o_0)+\mu b, \\ D &= 1-o_1, \end{aligned}$$

$$\begin{aligned} L &= \max\{C, D\}, \\ U &= \max\{A, C, D\}. \end{aligned} \tag{N3}$$

15.2.7.1 Proposition 3 (exact proposer-payoff projection for $N=3$) The exact recognized-proposer payoff projection is

$$\begin{aligned} \{U\}, & \quad \text{if } o_0 > 0; \\ [L, U], & \quad \text{if } o_0 = 0. \end{aligned} \tag{N3-PROJ}$$

A PBE exists for every primitive vector, every μ , and every value in the applicable projection.

Necessity: deviation lower bounds. The proposal $y=o_0, x_j=0$ gives at least C under every off-path ρ and completion: weak no gives C , while weak yes replaces the high-type continuation b by the current weak-only payoff one. Pooling at $y=o_1, x_j=0$ gives D . Hence every PBE value is at least L .

If $o_0 > 0$, there is also a genuine both-reject proposal $y=0, x_j=b$. Rejection then has probability one under every ρ ; the weak vote is relevant, equality prescribes yes, and the proposer receives A . Thus every PBE value is at least U when $o_0 > 0$. This argument is unavailable when $o_0 = 0$: at $y=0$ the low type accepts, so the proposal is separating rather than both-reject. An off-path assessment may set $\rho=0$ and complete the weak vote with no. The deviation then yields

$$(1-\mu)A + \mu b \leq C,$$

and does not force A .

Necessity: upper bound. Partition an on-support proposal, where $\rho=\mu$, into the same three regions. In a both-reject region (which exists only when $o_0 > 0$), weak yes gives at most A , while weak no gives $b \leq A$. At a separating proposal, if $\mu > 0$, weak yes gives

$$1 - x_j - (1-\mu)y \leq A,$$

and weak no gives at most C ; if $\mu=0$, the vote is a completion and the true payoff is at most $1-o_0=C$. Pooling gives at most D . Hence no PBE value exceeds U .

Sufficiency when $o_0=0$. Fix any V in $[L, U]$. At every distinct off-path separating proposal set $\rho=0$ and complete the weak vote with no. Such a deviation gives

$$(1-\mu)(1-y-x_j) + \mu b \leq C \leq V.$$

Every off-path pooling deviation gives at most $D \leq V$. There is no both-reject region. Put one of the following proposals on path with $\rho=\mu$:

1. if $V \leq A$, use $y=0, x_j=1-V$ and weak yes;
2. if $A < V \leq \max\{A, C\}$, necessarily $C > A$ and $\mu < 1$; use $y=0, x_j=(C-V)/(1-\mu) < b$ and weak no;
3. if $V > \max\{A, C\}$, the interval restrictions imply $V=D > \max\{A, C\}$; use pooling $y=o_1, x_j=0$.

The first proposal has $x_j \geq b$, the second has $x_j < b$, and the third makes the weak action irrelevant. Direct substitution gives proposer payoff V in each case. These on-path actions are sequentially rational, and the stated proposal-contingent off-path map bounds every deviation by V . Literal public-measurable $\kappa_0(h_2)$ selections complete every failed history; no active-H continuation is reached at $N=3$.

Sufficiency when $o_0 > 0$. The preceding upper bound is U . A candidate attaining whichever of A, C , or D equals U is, respectively, $(y, x_j)=(0, b)$, $(o_0, 0)$, or $(o_1, 0)$. Complete every distinct off-path both-reject proposal with its required threshold response, every separating proposal with $\rho=0$ and weak no, and every pooling proposal arbitrarily. Their payoffs are bounded by A, C , and D , respectively. This constructs value U and proves the singleton. QED.

15.2.7.2 Tie-break, endpoints, and mixing signatures For $o_0 > 0$, let $o_{\text{bar}}=(1-\mu)o_0 + \mu o_1$. Candidates A and C attain o_{bar} . If $D > \max\{A, C\}$, only the pooling value attains U and the tie-minimal H payoff is o_1 ; otherwise the tie-minimal H payoff is o_{bar} . A tied pooling signature survives only when it also ties that H payoff, which under $o_0 < o_1$ requires $\mu=1$.

For $o_0=0$, define $M=\max\{A,C\}$. At any equilibrium value $V \leq M$, a separating $y=0$ implementation gives H the lower bound μo_1 ; the tie-break retains only value- V signatures attaining that bound. Pooling signatures are then eliminated unless $\mu=1$, when their H payoff can tie. If $D > M$, the projection collapses to the endpoint $\{D\}$ and its tie-minimal implementation is pooling at $y=o_1, x_j=0$, with H payoff o_1 . These rules cover $\mu=0$, $\mu=1$, $\beta=1$, $V=L$, $V=U$, and all equality surfaces.

Proposal mixing is permitted only among support proposals that separately attain the same V and the same tie-minimal H payoff. It is not used to average unequal pure payoffs. For example, at $\beta=.5, o_0=0, o_1=.8, \mu=.9$, both separating proposals $(y, x_j)=(0,0)$ with weak no and $(0, .675)$ with weak yes give the lower endpoint $.325$ and H payoff $.72$; any mixture of those double-tied signatures is admissible under the common on-support belief $\rho=.9$. The exact projection in this example is $[\cdot325, .75]$, not the singleton $\{\cdot75\}$.

15.2.8 Constructive results for $N \geq 4$

15.2.8.1 Proposition 4 (a value-one PBE for every $N \geq 4$) For every admissible primitive vector, β , and μ , there is a PBE in which the proposer offers $y=0, X=0$, all weak nonproposers vote yes, type θ H votes yes exactly when $o_\theta=0$, and the current proposal passes. The proposer receives one in every type and H receives its outside payoff.

For $N \geq 5$, a unilateral weak no leaves both current H-yes and H-no branches passing, so every weak action is a completion. For $N=4$, a weak no after H no reaches weak-only C2; choose the admissible zero-vector C2-M-WO selection. The voter then compares zero with zero and yes is prescribed because the outcome is relevant. Every H type compares $y=0$ with o_θ . No proposer deviation can exceed one, and the construction already gives H its minimum type-specific payoff. This also supplies a complete off-path map: use all weak yes at every proposal (and zero-vector weak-only selections for the $N=4$ toggle histories).

Hence the global PBE correspondence is nonempty for every $N \geq 3$, including all boundary beliefs and $\beta=1$. Value one is an existence result, not a selection imposed on the rest of the correspondence.

15.2.8.2 Proposition 5 (proved $[0,1]$ proposer projections) The exact recognized-proposer payoff projection is $[0,1]$ in either of the following subclasses:

- (i) $N \geq 6$;
- (ii) N in $\{4,5\}$, $o_0 > 0$, and $\beta < 1$.

For any V in $[0,1]$, put on path $y=0$, gifts totaling $X=1-V$, and all weak yes. Current passage gives the proposer V in every type. Assign all weak no to every distinct proposal deviation.

In subclass (i), after all weak no and any unilateral weak deviation both H actions still fail the current ballot. Select a class-F active-H C2 outcome with weak vector zero and H payoff o_θ , and select the zero-vector weak-only C2 outcome. The weak actions are product-safe zero-payoff completions; H chooses no when $(\beta-1)o_\theta < 0$ and yes at equality. Every off-path proposal gives the proposer zero.

In subclass (ii), select the active-H C2 element with type payoff o_θ . Because $\beta < 1$ and $o_0 > 0$, both H types strictly choose immediate no after all weak no; the zero-vector weak-only selection then gives every weak player zero. Again every deviation gives the proposer zero. At $V=0$, the proposal and every deviation tie in proposer payoff and give H its outside payoff, so the primitive tie-break retains the construction. Feasibility gives the selection-free upper bound one, completing the proof.

Outside these subclasses, Proposition 2 remains exact; $[0,1]$ is not asserted.

15.2.9 The $o_0=0$ active-H continuation pathology

For N in $\{4,5\}$, the frozen active-H C2 interface changes discontinuously at $o_0=0$. Conditional on recognizing a Round-2 proposer, its payoff projection is

$$[D_A(nu), 1], \quad D_A(nu) = \max\{1-o_1, 1-nu\}, \quad (A0)$$

instead of the singleton $\{1\}$ obtained when $o_0 > 0$. Uniform recognition and nonnegative weak payoffs imply the assessment-independent, public-belief bound

$$E_{\text{nu}}[C^A_{\{W_j\}}(h_2)] \geq D_A(\text{nu}(h_2))/m \quad (A1)$$

for every weak identity j . The bound collapses only at the degenerate corner $\text{nu}=1, o_1=1$. If an on-path separating H-yes action reveals the low type, $\text{nu}(h_2)=0$, so (A1) equals $1/m$ and enters R1 as beta/m , exactly once.

Moreover, when $o_0=0$, type 0 compares immediate no payoff zero with a nonnegative active-H continuation on every H-yes failure profile. The response-at-equality rule makes it vote yes at equality. Therefore the all-failure punishment used in Proposition 5(ii) cannot make the low type opt out. At $\text{beta}=1, o_0>0$, the analogous outside-payoff C2 selection also makes every type vote yes at equality. These are genuine boundary changes, not permission to choose another C2 element or to replace (A0)–(A1) by a scalar.

Because $\text{nu}(h_2)$, the recognized Round-2 identity, full allocations, and off-path completions vary by history, this pathology enters (H-IC), (W-Y), and proposer optimality through the literal selection maps. No global continuation number, minimal coalition, or universal offer threshold survives.

15.2.10 Pre-recognition interface, existence, and invalidation

For each Round-1 weak identity i , let α_i be an assessment satisfying Proposition 2 and the proposal tie-break, including a proposal distribution σ_i . First integrate every type-by-identity coordinate and outcome distribution over that proposal draw as in (SIGMA). Before uniform recognition, for every player identity k and type θ ,

$$\begin{aligned} C1_{M,k}^{\alpha}(\theta|h_1) &= (1/m) \sum_i U_k^{\{i,\theta\}}(\alpha_i), \\ C1_{M,k}^{\alpha}(h_1) &= (1-\mu)C1_{M,k}^{\alpha}(0|h_1) + \mu C1_{M,k}^{\alpha}(1|h_1), \\ C1_M(h_1) &= \{\text{the resulting full vectors and distributions over all admissible} \\ &\quad \text{identity-indexed } \alpha=(\alpha_1, \dots, \alpha_m)\}. \end{aligned} \quad (C1-M)$$

The same $1/m$ and σ_i expectations apply to each type-conditional distribution over PR04–PR07, inclusion, opt-out, payment destinations, full public histories, and terminal/continuation signatures. The average retains every weak identity separately; it does not average weak players into a representative weak state. It also retains proposal mixing, ρ maps, complete ballot vectors, public-measurable action-specific C2 selections, and posteriors. Equal-posterior public histories remain distinct indices. The scalar proposer results in Sections 6–8 are implications of this object, not substitutes for it.

PBE existence is proved for every admissible primitive vector: Proposition 3 covers $N=3$ and Proposition 4 covers $N \geq 4$. Multiplicity is generally large. Exact scalar projections are proved only where stated; elsewhere the necessary-and-sufficient assessment fixed point is the result.

Any byte change to the Gate 0 bundle, either C2 interface, or the reviewed R2 batch invalidates this entire node and every descendant. A downstream node must consume the reviewed frozen replacement of this JSON interface. It may not rederive a continuation, silently select one, or apply another beta to the already Round-1-dated values exported here.

16 Collective entry

Entry is evaluated assessment by assessment. The full bargaining assessment is retained after the weak coalition forms or declines to form; a proposer payoff projection never determines collective value by itself.

16.1 Collective entry under unanimity

16.1.1 Status and frozen inputs

- Node: `entry_unanimity`.
- Solution concept: Perfect Bayesian equilibrium (PBE).

- Post-freeze status: exact interface accepted by two independent read-only reviews.
- Repair of rejected notation/documentation candidate: `sha256:efa5933a....`
- Validated population domain: every primitive-admissible $N \geq 3$ case.
- Frozen Gate 0 bundle: `sha256:6e28cb3f....`
- Frozen R1 batch: `sha256:f4ac7b89....`
- Frozen C1-U interface: `sha256:37aae1bf....`

This node applies the collective formation primitive to the complete frozen C1-U correspondence. It does not rederive a Round-1 or Round-2 strategy, change an assessment, select an equilibrium by a bound, or import a historical entry formula. All bargaining payoffs already use Round-1 units, so this node applies no further discount.

16.1.2 Primitive and information at entry

Write $m=N-1$. Nature draws the persistent type `theta` in $\{0,1\}$ before formation. The weak coalition does not observe `theta`. It makes one collective, all-or-nothing formation decision before the Round-1 recognition draw. If it forms, every weak state pays the same external sunk cost $\chi \geq 0$. The cost does not reduce the unit bargaining pie. If it does not form, every weak state receives zero, type `theta` of H receives `o_theta`, and the terminal outcome is nonformation.

The collective action is therefore the same in both realized types. It is not an individual participation vote. Under an identity-asymmetric bargaining assessment, formation can be collectively optimal even when one weak identity or one realized type obtains a negative net payoff.

Fix the public history and primitives, including the prior μ . Let $A_U(P)$ denote the nonempty set of complete C1-U assessments exported by the frozen unanimity interface. An element `alpha` in $A_U(P)$ includes one complete recognized-proposer PBE element for every possible Round-1 weak recognizer, with all proposal strategies, ballot strategies, beliefs, payments, continuation selections, type-by-player payoffs, and terminal-outcome kernels.

16.1.3 Dependency-safe integration

For recognized weak proposer i , the frozen interface first integrates over that proposer's proposal strategy `sigma_i`. For each type and weak identity, write the resulting coordinate as

```
E_sigma_i[U_Wk^i(theta;alpha_i)].
```

The corresponding H payoff and terminal-outcome kernel are integrated using the same `sigma_i` and the same assessment element. Uniform recognition is then applied, separately by type and player identity:

```
C1_W_alpha(theta,k)
= (1/m) sum_i E_sigma_i[U_Wk^i(theta;alpha_i)],
```

```
C1_H_alpha(theta)
= (1/m) sum_i E_sigma_i[U_H^i(theta;alpha_i)].
```

The type-conditional outcome kernel `D1_alpha(theta)` uses the same order and the same selection. Equal scalar payoffs do not permit payoff coordinates and outcome coordinates to be taken from different assessments.

Only after these type-by-identity C1 objects have been constructed define

```
T_W_alpha(theta) = sum_k C1_W_alpha(theta,k).
```

The gross value per weak state relevant to collective formation is

```
G_U(alpha,mu)
= [(1-mu) T_W_alpha(0) + mu T_W_alpha(1)] / m.
```

Thus the dependency-safe order is:

```
proposal support under sigma_i
-> E_sigma_i by type and identity
-> uniform recognition by type and identity
-> T_W_alpha(theta)
```

-> prior average
-> division by m
-> subtraction of external chi.

The operations happen to be linear, but retaining the intermediate objects is essential. Collapsing identities early would lose the realized payoff vector; splicing type or outcome coordinates could combine different alphas; and subtracting chi inside bargaining would change the game.

16.1.4 Assessment-level entry operator

Define the net collective value

$$Z_U(\alpha, \mu, \chi) = G_U(\alpha, \mu) - \chi.$$

The assessment-level action is

$$e_U(\alpha, \mu, \chi) = 1\{G_U(\alpha, \mu) \geq \chi\}.$$

Here $e_U=1$ means form. Equality forms exactly as specified in Gate 0.

16.1.4.1 Proposition 1 (entry action for a fixed assessment) For every complete α in $A_U(P)$, the unique collective entry action is formation if and only if $G_U(\alpha, \mu) \geq \chi$.

Proof. Conditional on formation, the collective primitive evaluates the average expected weak payoff as G_U ; every weak state then pays the external cost, so the average net payoff is $G_U - \chi$. Nonformation gives the weak coalition zero. A strictly positive net value selects formation, a strictly negative value selects nonformation, and the stated equality convention selects formation at zero. Because the coalition does not observe θ , the comparison integrates the two type rows using μ and produces one action for both realized types. No bargaining strategy is changed. QED.

16.1.5 Realized payoffs and outcomes

If α induces formation, realized type-by-identity weak payoffs are

$$u_{\text{entry_Wk}}(\theta; \alpha) = C1_W_alpha(\theta, k) - \chi,$$

and the type-conditional weak total is

$$\sum_k u_{\text{entry_Wk}}(\theta; \alpha) = T_W_alpha(\theta) - m\chi.$$

The hegemon and outcome coordinates are

$$\begin{aligned} u_{\text{entry_H}}(\theta; \alpha) &= C1_H_alpha(\theta), \\ D_{\text{entry_alpha}}(\theta) &= D1_alpha(\theta). \end{aligned}$$

The external cost is recorded as a payoff overlay for every weak identity. It does not change $D1_alpha$, payments, proposal feasibility, or H's payoff.

If α induces nonformation, then for both types

$$\begin{aligned} u_{\text{entry_Wk}}(\theta; \alpha) &= 0 \text{ for every } k, \\ u_{\text{entry_H}}(\theta; \alpha) &= o_{\theta}, \\ D_{\text{entry_alpha}}(\theta) &= \text{delta_nonformation}. \end{aligned}$$

The ex-ante outcome distribution is always formed only at the last step:

$$(1-\mu) D_{\text{entry_alpha}}(0) + \mu D_{\text{entry_alpha}}(1).$$

The gross counterfactual value G_U and the complete α remain recorded even when formation does not occur. They rationalize the entry action and specify every off-path bargaining history required by PBE.

16.1.6 Exact set-valued entry correspondence

Define the gross-value image

```
Gset_U(P,mu)
= {G_U(alpha,mu) : alpha in A_U(P)}.
```

Let $\text{Entry_U}(\alpha, \mu, \chi)$ be the complete record containing:

1. the unchanged assessment index and payload α ;
2. G_U , Z_U , and the form/no-form status;
3. both type rows of every weak identity's realized payoff;
4. both H type payoffs; and
5. the aligned type-conditional and ex-ante outcome distributions.

The entry correspondence is

```
E_U(P,mu,chi)
= {Entry_U(alpha,mu,chi) : alpha in A_U(P)}.
```

Its coarse status projection is

```
S_U(P,mu,chi)
= {form      : some alpha has G_U(alpha,mu)>=chi}
  union
  {no_form   : some alpha has G_U(alpha,mu)<chi}.
```

16.1.6.1 Proposition 2 (exactness and alpha preservation) E_U is the exact entry correspondence induced by frozen C1-U on its full $N \geq 3$ domain. It is nonempty. It cannot in general be replaced by its gross value set or status projection.

Proof. Frozen C1-U proves existence and attainment of at least one complete Round-1 PBE assessment for every primitive-admissible $N \geq 3$ case. Proposition 1 maps each such α to its unique sequentially rational collective entry action. If formation occurs, its continuation is exactly α ; if it does not occur, α supplies the complete off-path bargaining assessment. Conversely, any entry PBE induces a complete counterfactual C1-U assessment and its collective action must obey Proposition 1. This establishes equality, not merely inclusion.

Different α s can have the same G_U while assigning weak rents to different identities, giving H different type payoffs, or producing different outcome distributions. Removing α would therefore remove payoff- and outcome-relevant equilibrium multiplicity. QED.

16.1.7 Infimum, supremum, and endpoint attainment

The frozen C1-U bounds imply, for every α and type,

```
0 <= C1_W_alpha(theta,k),
0 <= T_W_alpha(theta) <= 1.
```

Consequently $Gset_U$ is a nonempty bounded subset of $[0, 1/m]$. Define

```
L_U(P,mu) = inf Gset_U(P,mu),
U_U(P,mu) = sup Gset_U(P,mu).
```

Both real endpoints exist. Boundedness does not imply that either endpoint is attained. This matters because the frozen continuation architecture contains nonclosed value boundaries.

The general C1-U interface does not supply closed-form numerical endpoint functions or a general upper-attainment theorem. For $N=3$, it also does not prove a general lower-attainment theorem. These facts are recorded as pending, not filled by the coarse interval $[0, 1/m]$.

To avoid any collision with the assessment set $A_U(P)$, denote endpoint attainment by the distinct indicators

```
a_U^- = 1 iff some alpha has G_U(alpha,mu)=L_U,
a_U^+ = 1 iff some alpha has G_U(alpha,mu)=U_U.
```

There is one useful exact special result. Frozen C1-U contains, for every $N \geq 4$, a complete assessment giving every weak identity zero in both types. Together with nonnegativity, this proves

$N \geq 4$: $L_U = 0$ and $a_U = 1$.

This is a property of an explicit frozen assessment, not an assumption that the entire interval is attainable.

16.1.8 Selection-free formation logic

16.1.8.1 Proposition 3 (complete endpoint logic) For every primitive-admissible case:

```
all_form
  iff  $L_U \geq \chi$ ;

possible_form
  iff  $\chi < U_U$ 
    or  $[\chi = U_U \text{ and } a_U = 1]$ ;

possible_no_form
  iff  $L_U < \chi$ ;

all_no_form
  iff  $U_U < \chi$ 
    or  $[\chi = U_U \text{ and } a_U = 0]$ .
```

Proof. All assessments form exactly when every element g of G_{set_U} satisfies $g \geq \chi$, which is equivalent to $\inf G_{set_U} \geq \chi$. Some assessment does not form exactly when some $g < \chi$, equivalent to $\inf G_{set_U} < \chi$.

If $\chi < U_U$, the definition of supremum guarantees an element strictly above χ , hence formation is possible. If $\chi > U_U$, no element reaches χ . At $\chi = U_U$, formation is possible exactly when the supremum belongs to the set. Negating that statement gives the all-no condition. QED.

Two equality cases deserve emphasis.

- At $\chi = L_U$, every assessment forms, whether or not the infimum is attained. Every value is at least its infimum, and equality forms.
- At $\chi = U_U$, some assessment forms if and only if the supremum is attained. If it is not attained, every value is strictly below the cost and every assessment chooses nonformation.

If $L_U = U_U = \chi$, nonemptiness implies that the value set is the attained singleton $\{\chi\}$. Every assessment forms and no assessment chooses nonformation.

16.1.9 Direct selection-free bounds and cost regions

The type-specific total bound yields directly

```
0 ≤ G_U(alpha,mu) ≤ 1/m,
-χ ≤ G_U(alpha,mu) - χ ≤ 1/m - χ.
```

When formation occurs, each weak identity and the total weak coalition obey

```
-χ ≤ C1_W_alpha(theta,k) - χ ≤ 1 - χ,
-m*χ ≤ T_W_alpha(theta) - m*χ ≤ 1 - m*χ.
```

These are payoff bounds, not an assertion that their endpoints are feasible. They give the following selection-free conclusions:

1. $\chi = 0$: every assessment forms, including any zero-value assessment by equality.
2. $\chi > 1/m$: every assessment chooses nonformation.
3. $\chi = 1/m$: formation is possible only if a complete alpha attains the bound $G_U = 1/m$; otherwise every assessment chooses nonformation.

4. $0 < \chi < 1/m$: the coarse bounds alone do not classify the correspondence.
5. For $N \geq 4$ and every $\chi > 0$, the attained zero-value assessment chooses nonformation. Thus nonformation is possible and all-form is false. Whether formation is also possible depends on the upper endpoint and its attainment.

No zero-value assessment is asserted for general $N=3$ primitives. A consumer must use the exact C1 image or a separately proved sharper result.

16.1.10 Primitive boundaries and domain

The operator uses the full common C1-U existence domain:

```
integer N>=3,
mu in [0,1],
0<=o_0<o_1<=y_bar<=1,
beta in (0,1],
chi>=0.
```

It includes $N=3$, degenerate priors, $o_0=0$, $o_1=y_{\text{bar}}$, $y_{\text{bar}}=1$, $\beta=1$, $\chi=0$, $\chi=1/m$, and costs above $1/m$. Degenerate priors retain both type coordinates because the complete assessment and counterfactual payoffs remain part of the interface. The entry operator does not add a new use of **beta**; it consumes C1 in native Round-1 units.

16.1.11 Sufficiency boundary

The frozen C1-U payload is sufficient for:

- the exact alpha-preserving entry correspondence;
- every assessment-level entry decision;
- realized type-by-identity weak and H payoffs;
- aligned outcome distributions;
- exact endpoint-conditional formation logic; and
- the general bounds and $N \geq 4$ zero lower endpoint proved above.

It is not sufficient for a general closed-form formula for L_U , U_U , or upper attainment. This does not block the exact set-valued entry operator. Any later claim of numeric endpoint functions, an attained interval $[0, 1/m]$, or a bound-selected alpha would require reopening or further solving C1-U. None is claimed here.

16.1.12 Mechanical validation executed

The companion R verifier completed with **37/37 PASS**. It generated 72 full-assessment fixture rows, 48 realized type-by-identity payoff rows, and seven endpoint-logic cases; exercised three dependency mutation guards; and confirmed all 27 protected hashes. Specifically, it checked:

- all three exact dependency hashes and mutation failures for C1-U and the R1 batch freeze;
- every protected file in the 27-file manifest;
- synthetic full-vector alphas with all recognizers, both types, every weak identity, proposal-support mixing, H coordinates, and outcome kernels;
- **E_sigma**, recognition, type, and weak-average integration against manually computed values;
- identity asymmetry and two distinct alphas with equal gross value;
- external-cost accounting, equality formation, nonformation payoffs, and aligned H/outcome coordinates;
- exact finite-correspondence preservation;
- symbolic attained and unattained endpoint cases;
- $N \geq 3$ and every listed primitive boundary; and
- negative mutations that scalarize identities, change averaging, put cost in the pie, default endpoint attainment, or rediscout C1.

Numerical fixtures check the operator. They do not replace Propositions 1–3.

16.1.13 Claim ledger

- **Assessment-level formation operator.**

- **Status:** proved
- **Type-by-identity realized payoff and outcome map.**
 - **Status:** proved
- **Exact alpha-preserving entry correspondence.**
 - **Status:** proved
- **Complete infimum/supremum boundary logic.**
 - **Status:** proved
- **General gross bound $[0, 1/m]$.**
 - **Status:** proved from frozen C1-U
- **$N \geq 4$ lower endpoint zero and attained.**
 - **Status:** proved from frozen C1-U
- **General $N=3$ endpoint values/attainment.**
 - **Status:** pending; not claimed
- **General upper endpoint value/attainment.**
 - **Status:** pending; not claimed
- **Closed-form entry regions.**
 - **Status:** pending; not claimed
- **Historical entry formulas as inputs.**
 - **Status:** rejected
- **Mechanical fixture audit.**
 - **Status:** checked numerically: 37/37 PASS
- **Independent read-only reviews.**
 - **Status:** PASS on the exact repaired hash

Any byte change to Gate 0, the R1 batch, or C1-U invalidates this frozen interface. The independent review bundle later marked the exact node PASS.

16.2 Collective entry under majority

16.2.1 Scope and frozen input

This node lifts the frozen majority Round-1 correspondence into the collective entry decision. It consumes the exact artifacts

Gate 0 bundle	sha256:6e28cb3f...
R1 batch	sha256:f4ac7b89...
C1 majority	sha256:21c3a9dd...

No R1 or C2 object is rederived, repaired, or scalarized. The admissible population is $N=m+1 \geq 3$. Every C1 payoff is already in Round-1 units, so this node applies no further discount.

The weak states face one collective, all-or-nothing formation decision before they observe **theta**. If they form, every weak identity pays the external sunk cost **chi** ≥ 0 . If they do not form, every weak state receives zero and H type **theta** receives **o_theta**. Equality selects formation. This is a collective formation device evaluated by the average weak payoff, not a set of unilateral identity-level participation decisions.

16.2.2 The assessment-level value operator

Fix a complete admissible C1-M assessment

$\alpha = (\alpha_1, \dots, \alpha_m)$ in $A_M(h_1)$.

The coordinate α_i is the complete assessment conditional on weak identity i being recognized. It includes σ_i , proposal-contingent beliefs, all ballot actions and completions, posteriors, action-specific C2 selections, type-by-identity payoffs, gifts, inclusion and opt-out status, full histories, and outcome distributions.

For each i , first integrate every player and type coordinate over σ_i . Then average over uniform recognition:

$$C1_M, k^\alpha(\theta|h_1) = (1/m) \sum_i U_k^{\{i, \theta\}}(\alpha_i). \quad (1)$$

For each named weak identity j , integrate the true type distribution:

$$\begin{aligned}
C1_M, Wj^{\alpha}(\mu|h1) \\
= (1-\mu) C1_M, Wj^{\alpha}(0|h1) \\
+ \mu C1_M, Wj^{\alpha}(1|h1).
\end{aligned} \tag{2}$$

Finally average across the m weak identities:

$$\begin{aligned}
V_W^M(\alpha, \mu) \\
= (1/m) \sum_j C1_M, Wj^{\alpha}(\mu|h1).
\end{aligned} \tag{3}$$

Equivalently, define the type-specific aggregate

$$T_W^M(\alpha, \theta) = \sum_j C1_M, Wj^{\alpha}(\theta|h1).$$

Then

$$\begin{aligned}
V_W^M(\alpha, \mu) \\
= [(1-\mu)T_W^M(\alpha, 0) + \mu T_W^M(\alpha, 1)]/m.
\end{aligned} \tag{4}$$

Equations (3) and (4) coincide by finite-sum linearity. The canonical computational order is $E_{\sigma} \rightarrow \text{recognition} \rightarrow \text{type} \rightarrow \text{cross-weak}$. This order preserves every named identity through all upstream expectations. It never averages across distinct assessments.

16.2.3 Exact entry lift

For the fixed complete assessment α , define

$$N_W^M(\alpha, \mu, \chi) = V_W^M(\alpha, \mu) - \chi.$$

The collective rule is

$$\text{form}(\alpha) \text{ iff } V_W^M(\alpha, \mu) \geq \chi. \tag{ENTRY-M}$$

At equality the institution forms. Conditional on formation, every named weak identity and H receive

$$\begin{aligned}
E_M, Wj^{\alpha}(\theta) &= C1_M, Wj^{\alpha}(\theta|h1) - \chi, \\
E_M, H^{\alpha}(\theta) &= C1_M, H^{\alpha}(\theta|h1).
\end{aligned} \tag{5}$$

The complete type-conditional $C1$ outcome distribution is inherited without change and receives only two new coordinates: $\text{entry_status} = \text{formed}$ and an external cost ledger recording χ for each weak identity. The cost does not reduce the institutional pie, alter a gift, transfer to H , enter a ballot IC , or receive β . Because the decision criterion is collective, a particular identity's net payoff in (5) may be negative; it is retained rather than truncated or converted into an unstated individual veto.

Conditional on nonformation,

$$\begin{aligned}
E_M, Wj(\theta) &= 0 \text{ for every } j, \\
E_M, H(\theta) &= o_{\theta}.
\end{aligned} \tag{6}$$

No proposal, ballot, $R1$ continuation, or $R2$ continuation is realized.

16.2.3.1 Proposition 1 (exact entry correspondence) Let $A_M(h1)$ be the full frozen $C1$ - M assessment correspondence. The entry correspondence consists exactly of one object for every whole α in $A_M(h1)$: retain α , calculate (1)–(4), apply (ENTRY-M), and attach the payoffs and outcomes in (5) or (6).

Proof. Any PBE of the entry-augmented game specifies a complete $C1$ - M continuation assessment following formation, including when formation is off path. Sequential rationality of the collective decision requires formation exactly when its average continuation payoff net of χ is nonnegative, with formation at equality. This proves necessity. Conversely, any frozen $C1$ - M assessment is sequentially rational in its continuation game. Pairing it with the action prescribed by (ENTRY-M) makes the collective entry action optimal, and (5)–(6) are the primitive payoff map. This proves sufficiency. QED.

The proposition does not select an assessment by its bound, proposer payoff, or entry status. It preserves the entire α on both sides of the entry comparison.

16.2.4 Selection-free envelope and exact cost regions

Define the nonempty value set and its endpoints

```
S_M(mu) = {V_W^M(alpha, mu) : alpha in A_M(h1)},
v_M^-   = inf S_M(mu),
v_M^+   = sup S_M(mu).
```

Let a_M^- and a_M^+ indicate whether some complete assessment attains the respective endpoint. Attainment is recorded only when proved; it is not inferred from the existence of a supremum.

16.2.4.1 Proposition 2 (resource envelope) For every admissible primitive vector,

$$0 \leq v_M^- \leq v_M^+ \leq 1/m. \quad (7)$$

For every $N \geq 4$,

$$v_M^+ = 1/m \text{ and } a_M^+ = \text{true}. \quad (8)$$

No universal exact lower endpoint or lower-attainment claim follows from the frozen C1-M interface. At $N=3$, neither endpoint may be read from the recognized-proposer projection.

Proof. On PR04, aggregate weak gross payoff is $1-y$; on PR06 it is one. On PR05 and PR07, the selected C2 allocation has aggregate weak payoff between zero and one in native R2 units and is multiplied by $\beta \leq 1$ exactly once. Thus every realized R1 branch has aggregate weak gross payoff in $[0, 1]$. Proposal mixing, ballot integration, uniform recognition, and type integration preserve the interval. Division by m proves (7). For $N \geq 4$, the frozen C1-M value-one construction gives the recognized proposer one, every other weak state zero, and current passage in every type. Its aggregate weak payoff is one before and after recognition, so it attains $1/m$. Combined with (7), this proves (8). QED.

The exact assessment-level classification is

- **all assessments form.**
 - Necessary and sufficient condition: $\chi \leq v_M^-$
- **formation is possible.**
 - Necessary and sufficient condition: $\chi < v_M^+$, or $\chi = v_M^+$ and $a_M^+ = \text{true}$
- **nonformation is possible.**
 - Necessary and sufficient condition: $\chi > v_M^-$
- **all assessments choose no.**
 - Necessary and sufficient condition: $\chi > v_M^+$, or $\chi = v_M^+$ and $a_M^+ = \text{false}$

At $\chi = v_M^-$, all assessments form regardless of lower-endpoint attainment: every element of the set is at least its infimum and equality forms. At $\chi = v_M^+$, by contrast, formation requires actual upper-endpoint attainment. This is the only nonattainment-sensitive boundary in the four classifications.

Immediate selection-free implications are:

- $\chi = 0$: every assessment forms;
- $\chi > 1/m$: every assessment chooses no;
- $N \geq 4$, $\chi = 1/m$: formation is possible through the attained value-one assessment.

No stronger common assessment selection is supplied here.

16.2.5 Why proposer projections cannot classify entry

The scalar results frozen in C1-M concern the recognized proposer's payoff, not (3). The distinction survives every proposer and H tie-break.

Consider the frozen counterexample

$N=3$, $\beta=.5$, $o_0=0$, $o_1=.8$, $\mu=.9$.

The recognized-proposer projection is $[\cdot 325, \cdot 75]$. At its lower endpoint, the two separating proposals

```
(y,x)=(0,0),      with the other weak state voting no;
(y,x)=(0,.675),   with the other weak state voting yes
```

both give the proposer .325 and H .72, so the frozen tie-break retains both and permits their mixture. Under the first signature, type 0 yields a current agreement with aggregate weak payoff one, while type 1 reaches the weak-only continuation with discounted aggregate weak payoff .5. Therefore

```
T_W ex ante=.1(1)+.9(.5)=.55,  V_W=.55/2=.275.
```

Under the second signature, current passage allocates the full institutional pie to weak states in both types, so $V_W=1/2$. Mixing the double-tied signatures preserves proposer payoff .325 and H payoff .72 while spanning collective values [.275, .5]. Hence even a fixed proposer payoff plus the H tie-break does not identify entry.

The same accounting explains why the proved $[0,1]$ proposer projections do not yield $[0,1/m]$ collective-value projections. Their displayed construction sets $y=0$, gives gifts totaling $1-V$, and passes currently. The proposer and the other weak states jointly receive one for every V , so every member of that proposer-payoff continuum maps to the same collective value $1/m$.

16.2.6 Status and invalidation

The exact artifact was subsequently accepted by two independent read-only reviews. Its R verifier checks the exact frozen bytes and transitive components, but does not require the historical R1-batch readiness snapshot to remain unchanged after the entry nodes have legitimately started. It also checks asymmetric identity and proposal-mixing fixtures, operation order, equality, external costs, nonformation, endpoint logic including an unattained supremum, the resource envelope, the $N \geq 4$ upper endpoint, and the $N=3$ negative shortcut.

Any byte change to Gate 0, the R1 batch, C1-M, or a frozen transitive component invalidates this node and every descendant. Two independent read-only reviews subsequently accepted the exact candidate hash.

17 Institutional comparison

The exact comparison object is the Cartesian product of whole entry assessments under the two fixed rules. There is no primitive that couples, rank-matches, or jointly selects assessments across rules.

17.1 Institutional comparison on the full product of entry assessments

17.1.1 Scope and exact frozen inputs

This node compares the two fixed institutional rules only after the entry correspondences have been independently closed. It consumes literally:

```
Entry batch      sha256:8817a9c5...
Entry unanimity  sha256:05e39ad2...
Entry majority   sha256:4eb343ce...
```

The common PBE-existence domain is $N=m+1 \geq 3$, μ in $[0,1]$, $0 \leq o_0 < o_1 \leq y_{\text{bar}} \leq 1$, β in $(0,1]$, and $\chi \geq 0$. C1 payoffs are already expressed in Round-1 units. This comparison applies neither β nor the entry cost a second time.

Nothing here rederives R1, C2, or entry. Unanimity and majority remain fixed counterfactual rules. The model has no endogenous rule-choice or signaling stage.

17.1.2 Why the comparison is a Cartesian product

Let $A_U(P)$ and $A_M(P)$ denote the exact complete assessment sets carried by the two approved entry interfaces at primitive-history index P . The game contains no primitive that correlates equilibrium selection across counterfactual institutions: no common random device, rank matching, shared off-path belief, or rule-choice stage. The exact comparison index is therefore

$$A_U(P) \times A_M(P). \quad (1)$$

Every element of (1) retains both assessments whole. In particular, it keeps all type-by-identity weak payoffs, both H type coordinates, strategies, beliefs, continuation selections, cost ledgers, and aligned outcome kernels. Equal scalar values do not merge assessments. Conversely, matching two assessments by scalar rank or proposer payoff would add a coupling absent from the game.

17.1.2.1 Proposition 1 (exact comparison correspondence) For fixed P, mu, and chi, the institutional-comparison correspondence is

$$\begin{aligned} C(P, \mu, \chi) \\ = \{ \text{Comp}(\alpha_U, \alpha_M; \mu, \chi) : \\ (\alpha_U, \alpha_M) \text{ in } A_U(P) \times A_M(P) \}. \end{aligned} \quad (2)$$

Proof. Each coordinate of a comparison must be a complete entry PBE assessment of its own fixed-rule game, which proves that any admissible pair lies in the right side of (2). In the absence of a cross-rule selection primitive, no restriction links an admissible U assessment to a proper subset of admissible M assessments. Thus every cross-pair is admissible. The comparison operator below only reads and retains the coordinates; it changes neither assessment. QED.

17.1.3 Assessment-pair payoff and outcome operator

Fix one whole pair (α_U, α_M) . For $R \text{ in } \{U, M\}$, import the collective gross value

$$G_R = G_R(\alpha_R, \mu)$$

from the corresponding entry element and define

$$e_R = 1_{\{G_R \geq \chi\}}. \quad (3)$$

Equality forms. The realized collective net payoff per weak state is

$$W_R = e_R * (G_R - \chi). \quad (4)$$

It is zero if the institution does not form. For each H type,

$$\begin{aligned} H_R(\theta) \\ = e_R * C1_R, H^{\alpha_R}(\theta) + (1 - e_R) * o_{\theta}, \end{aligned} \quad (5)$$

$$H_R(\mu) = (1 - \mu) H_R(0) + \mu H_R(1). \quad (6)$$

The prior in (6) is the true mu, not a proposal-contingent ballot belief. Define the U-minus-M differences

$$\begin{aligned} \Delta_W &= W_U - W_M, \\ \Delta_H(\theta) &= H_U(\theta) - H_M(\theta), \\ \Delta_H(\mu) &= H_U(\mu) - H_M(\mu). \end{aligned} \quad (7)$$

The formation pattern is both, U_only, M_only, or neither according to $(e_U, e_M) = (1, 1), (1, 0), (0, 1), (0, 0)$. The comparison record retains the two complete realized outcome distributions in addition to this label and the payoff summaries.

17.1.4 Exact pairwise formation-set comparison

For a fixed assessment, define its cost-indexed formation set

$$F_R(\alpha_R) = \{\chi \geq 0 : e_R = 1\} = [0, G_R]. \quad (8)$$

17.1.4.1 Proposition 2 (pairwise threshold and nesting) For every complete cross-pair,

$$\begin{aligned} F_U \text{ subseq } F_M & \text{ iff } G_U \leq G_M, \\ F_M \text{ subseq } F_U & \text{ iff } G_M \leq G_U, \\ F_U = F_M & \text{ iff } G_U = G_M. \end{aligned} \quad (9)$$

The corresponding inclusion is strict iff the displayed threshold inequality is strict.

At a fixed cost,

$$\begin{aligned} \text{both:} & \quad 0 \leq \chi \leq \min\{G_U, G_M\}, \\ \text{U_only:} & \quad G_M < \chi \leq G_U, \\ \text{M_only:} & \quad G_U < \chi \leq G_M, \\ \text{neither:} & \quad \chi > \max\{G_U, G_M\}. \end{aligned} \tag{10}$$

The U_only region is nonempty exactly when $G_U > G_M$; the M_only region is nonempty exactly when $G_M > G_U$.

Proof. Equation (8) follows directly from (3) and equality formation. Inclusion of two closed intervals with common lower endpoint zero is equivalent to ordering their upper endpoints, which proves (9). Intersecting the two weak inequalities in (3), or one weak and one strict inequality for an exclusive pattern, gives (10). QED.

The weak inequality on the forming side of each exclusive region matters. At $\chi = G_R$, rule R forms with zero net weak payoff.

17.1.5 Selection-free pattern existence and endpoint attainment

For each rule define the nonempty value set and endpoints

$$\begin{aligned} S_R(P, \mu) &= \{G_R(\alpha_R, \mu) : \alpha_R \text{ in } A_R(P)\}, \\ l_R &= \inf S_R, \quad u_R = \sup S_R. \end{aligned}$$

Let $a_R^+ = 1$ exactly when an assessment attains u_R . Then

$$\begin{aligned} \text{PF}_R(\chi) &:= [\chi < u_R] \text{ or } [\chi = u_R \text{ and } a_R^+ = 1], \\ \text{PN}_R(\chi) &:= [l_R < \chi]. \end{aligned} \tag{11}$$

Here PF means that formation is possible and PN means that nonformation is possible. The upper-attainment qualifier in (11) is indispensable: when $\chi = u_R$ and the supremum is not attained, every assessment has $G_R < \chi$. At the lower endpoint, PN_R is false regardless of lower attainment because no value can lie strictly below its infimum.

17.1.5.1 Proposition 3 (exact status-pattern existence) At a fixed cost,

- **both.**
 - Exists if and only if: PF_U and PF_M
- **U_only.**
 - Exists if and only if: PF_U and PN_M
- **M_only.**
 - Exists if and only if: PN_U and PF_M
- **neither.**
 - Exists if and only if: PN_U and PN_M

Proof. The rule-specific status projections are exactly the sets generated by (11). Proposition 1 makes the joint projection their Cartesian product. Each row is therefore the conjunction of the two required marginal statuses. QED.

The phrase “formation set” can now refer to three different objects. They must not be conflated. Within each rule, define the costs at which at least one assessment forms and the costs at which every assessment forms:

$$\begin{aligned} K_R^{\text{exists}} &= \text{union}_{\{\alpha_R\}} F_R(\alpha_R), \\ K_R^{\text{forall}} &= \text{intersection}_{\{\alpha_R\}} F_R(\alpha_R). \end{aligned} \tag{12}$$

17.1.5.2 Proposition 4 (possible-cost and guaranteed-cost projections) The exact projections are

$$\begin{aligned} K_R^{\text{exists}} &= [0, u_R] \quad \text{if } a_R^+ = 1, \\ &\quad [0, u_R) \quad \text{if } a_R^+ = 0, \\ K_R^{\text{forall}} &= [0, l_R]. \end{aligned} \tag{13}$$

Consequently,

$$\begin{aligned} K_U^{\text{exists}} \text{ subseq } K_M^{\text{exists}} \\ \text{iff } u_U < u_M, \\ \text{or } u_U = u_M \text{ and } [a_U^+ = 0 \text{ or } a_M^+ = 1]; \end{aligned} \quad (14)$$

$$K_U^{\text{forall}} \text{ subseq } K_M^{\text{forall}} \text{ iff } l_U \leq l_M. \quad (15)$$

The reverse conditions exchange U and M. The possible-cost sets are equal iff $u_U = u_M$ and $a_U^+ = a_M^+$; inclusion is strict when inclusion holds and this equality condition fails. The guaranteed-cost sets are equal iff $l_U = l_M$ and inclusion is strict iff the corresponding endpoint inequality is strict.

Proof. The union in (12) contains every cost strictly below the supremum and contains the supremum exactly when it is attained. The intersection contains every cost weakly below the infimum, including the infimum whether or not some assessment attains it. Inclusion and equality of the resulting half-intervals give (14)–(15) and their reverse statements. QED.

These union/intersection projections are not the condition that every U assessment's interval be nested in every M assessment's interval. That third, strictly cross-assessment object is next.

17.1.5.3 Proposition 5 (universal cross-assessment nesting) Across every complete cross-pair,

$$\begin{aligned} F_U(\alpha_U) \text{ subseq } F_M(\alpha_M) \text{ for all pairs} \\ \text{iff } u_U \leq l_M, \end{aligned} \quad (16)$$

$$\begin{aligned} F_M(\alpha_M) \text{ subseq } F_U(\alpha_U) \text{ for all pairs} \\ \text{iff } u_M \leq l_U. \end{aligned} \quad (17)$$

Let a_R^- record lower-endpoint attainment. Strict inclusion for **every** cross-pair has the endpoint-sensitive characterization

$$\begin{aligned} F_U(\alpha_U) \text{ strict-subset } F_M(\alpha_M) \text{ for all pairs} \\ \text{iff } u_U < l_M, \\ \text{or } u_U = l_M \text{ and not } [a_U^+ = 1 \text{ and } a_M^- = 1], \end{aligned}$$

with the symmetric condition for strict M-in-U. When the two separating endpoints coincide, equality of one pair is possible exactly if both relevant endpoints are attained. All cross-pair formation sets are equal exactly when both value sets are the same singleton: $l_U = u_U = l_M = u_M$.

Proof. By Proposition 2, the left side of (16) is the statement that every element of S_U is no larger than every element of S_M . This holds exactly when $\sup S_U \leq \inf S_M$. The proof of (17) is symmetric. If all cross-pair thresholds are equal, every element of each nonempty set equals the same number; the converse is immediate. QED.

These are exact endpoint characterizations, not numerical rankings. The frozen interfaces do not generally identify u_U or l_M ; at $N=3$ they do not generally identify the reverse pair either. Accordingly, no general U-in-M conclusion is asserted. Same-rank or endpoint-matched checks cannot replace the universal cross-product conditions.

17.1.6 H's outside-option floor and conditional rankings

17.1.6.1 Lemma 1 (H participation floor inherited from C1) For every complete frozen C1 assessment, rule, and type,

$$C1_R, H^{\alpha_R}(\theta) \geq o_{\theta}. \quad (18)$$

Proof. At every H ballot information set under either rule, voting **no** irreversibly opts H out. For every simultaneous weak-vote realization this action gives type θ exactly o_{θ} , without discount. The PBE ballot fixed point makes H's equilibrium behavioral action a best response against the distribution of weak ballots, so its conditional expected payoff is at least the payoff from the available **no** action. This argument does not let H observe the sealed weak ballots and does not reduce relevance to the quota. Integrating the conditional inequality over the equilibrium proposal mixture and uniform weak recognition preserves it, proving (18). QED.

Equations (5) and (18) imply $H_R(\theta) \geq o_\theta$ at entry. They yield the following exact conditional comparisons:

- **neither.**
 - **Weak U-minus-M:** $\Delta_W = 0$
 - **H U-minus-M:** $\Delta_H(\theta) = 0$ for both types
- **U_only.**
 - **Weak U-minus-M:** $\Delta_W = G_U - \chi \geq 0$
 - **H U-minus-M:** $\Delta_H(\theta) = C1_U, H(\theta) - o_\theta \geq 0$
- **M_only.**
 - **Weak U-minus-M:** $\Delta_W = -(G_M - \chi) \leq 0$
 - **H U-minus-M:** $\Delta_H(\theta) = o_\theta - C1_M, H(\theta) \leq 0$
- **both.**
 - **Weak U-minus-M:** $\Delta_W = G_U - G_M$
 - **H U-minus-M:** sign of $C1_U, H(\theta) - C1_M, H(\theta)$ is undetermined

Typewise inequalities imply their true-prior ex ante counterparts. These are pair-conditional identities and bounds. They do not select an assessment or establish unconditional institutional dominance.

The floor is attained whenever nonformation is possible. If $PN_R(\chi)$ holds, there is a complete assessment with $G_R < \chi$; its entry action is no formation and (5) gives H exactly (o_0, o_1) . If all assessments choose no, every entry element attains that vector. In particular, $\chi > 1_R$ is the exact condition for at least one no-form element and hence for this direct floor-attainment argument.

There are two useful equivalent decompositions. First, over the entire cost axis,

$$W_R(\chi) = \max\{G_R - \chi, 0\}.$$

Thus $G_U \leq G_M$ if and only if $W_U(\chi) \leq W_M(\chi)$ for every $\chi \geq 0$; the curves coincide for every cost iff $G_U = G_M$. Second, define H's nonnegative type-specific rent

$$r_R(\theta) = C1_R, H(\theta) - o_\theta \geq 0.$$

Then

$$\begin{aligned} H_R(\theta) &= o_\theta + e_R * r_R(\theta), \\ \Delta_H(\theta) &= e_U * r_U(\theta) - e_M * r_M(\theta). \end{aligned} \tag{19}$$

This makes clear why the H rankings in the exclusive patterns are weak, not strict: the forming institution can give H exactly its outside option.

The common resource envelope gives the selection-free weak bounds

$$\begin{aligned} 0 &\leq W_R \leq \max\{1/m - \chi, 0\}, \\ |\Delta_W| &\leq \max\{1/m - \chi, 0\}. \end{aligned} \tag{20}$$

Equal formation status or equal payoff does not imply equal payments, inclusion, opt-out histories, beliefs, or outcome distributions.

17.1.7 Boundary and special-endpoint implications

Both frozen entry interfaces prove $0 \leq G_R \leq 1/m$. Therefore:

- At $\chi = 0$, every assessment forms under both rules, including at $G_R = 0$. Every comparison pair has pattern **both**.
- At $\chi > 1/m$, no assessment forms under either rule. Every pair has pattern **neither**, every weak payoff is zero, and each H type receives o_θ .
- At $\chi = 1/m$, formation still occurs for any assessment attaining the upper resource bound; an unattained supremum is not enough.

For every $N \geq 4$, the approved U entry interface proves $1_U = 0$ with an attaining assessment, while the approved M entry interface proves $u_M = 1/m$ with an attaining assessment.

These endpoints imply two within-rule projection results:

$$\begin{aligned} K_U^{\text{exists}} &= [0, 1/m], \text{ so } K_U^{\text{exists}} \subseteq K_M^{\text{exists}}; \\ K_U^{\text{forall}} &= \{0\}, \text{ so } K_U^{\text{forall}} \subseteq K_M^{\text{forall}}. \end{aligned} \quad (21)$$

The first inclusion may be equality or strict depending on u_U and upper attainment. The second may be equality or strict depending on l_M . Neither statement is the universal cross-assessment condition in Proposition 5.

17.1.7.1 Corollary 1 (existence of an M-only pair for $N \geq 4$) For every $N \geq 4$ and every $0 < \chi \leq 1/m$, there exists a complete cross-pair with pattern M_{only} .

Proof. Pair the attained U assessment with $G_U = 0$ and the attained M assessment with $G_M = 1/m$. Positive cost makes U choose no. Majority forms because $\chi \leq 1/m$, including equality at $\chi = 1/m$. Proposition 1 admits this cross-pair. QED.

The frozen M value-one construction used here gives H its type-specific outside payoff even though majority forms. U nonformation does the same. Therefore this actual extreme pair has

$$\begin{aligned} \Delta_H(\theta) &= 0 \text{ for both types,} \\ \Delta_W &= -(1/m - \chi). \end{aligned}$$

The weak difference is strictly negative for $0 < \chi < 1/m$ and equals zero at $\chi = 1/m$, when majority forms at collective indifference. Thus even the guaranteed M_{only} example does not imply a strict H advantage.

Because $u_M = 1/m > 0 = l_U$, the corollary also rules out universal cross-assessment $M_{\text{in-}U}$ nesting for $N \geq 4$. Universal cross-assessment $U_{\text{in-}M}$ remains pending because it requires $u_U \leq l_M$. The corollary does **not** show that every pair is M-only or that the model selects majority. At $N = 3$, neither special endpoint is imported; only the general pairwise and endpoint-conditional results apply.

17.1.8 Verification and status

The R verifier checks exact dependency bytes; the complete-product rather than coupled comparison; exhaustive nonempty finite value subsets and cross-pairs; attained and unattained endpoint boundaries; randomized full assessment fixtures with asymmetric identities, proposal mixtures, type-specific H coordinates, and retained outcome labels; the $N = 3$ and $N \geq 4$ boundaries; false-nesting and proposer-projection mutations; three dependency mutations; and all 27 protected hashes.

The machine checks validate algebra and guards; they do not replace the proofs above. The exact artifact was subsequently accepted by two independent read-only reviews. Its frozen review bundle, rather than this implementation snapshot, authorizes the nonmigrating survival inventory to consume it.

18 Nonmigrating survival inventory

The claim-level matrix compares historical statements with the frozen current objects only after the new derivation is complete. It is a provenance and migration-warning artifact, not a source of equilibrium results and not an authorization to edit a manuscript.

18.1 v6 survival matrix: claim-level, nonmigratory classification

This frozen inventory records how the completed rederivation bears on historical claims; it neither selects an equilibrium nor authorizes manuscript migration.

18.1.1 Scope and frozen dependencies

This node classifies historical manuscript claims against the approved current correspondences. It is a provenance matrix, not a new derivation and not a manuscript rewrite. Its only formal comparison dependencies are:

Institutional comparison
sha256: cab69c5d...

Independent comparison review bundle
sha256: 0acd9648...

The current object is PBE under local outcome-signature relevance, yes at payoff-relevant equality, and explicit full-profile completion when a vote has zero outcome relevance. It is not described by an imported historical label. The comparison domain is the full primitive-admissible $N \geq 3$ domain exported by the frozen entry interfaces, with no cross-rule assessment selection.

Historical material was read only to locate claims. The four inventories and their frozen hashes are `v5 manuscript` (1b0e420...), `v6 manuscript` (131cc235...), `prior derivation` (8fbb7ed...), and `AGENTS.md` (2164d247...). None supplies current proof evidence.

18.1.2 Machine-readable result

The authoritative CSV contains 53 source-claim-granular rows and exactly 11 columns. Every row records its historical path and inclusive line locator, plain-language historical claim, one allowed status, precise current replacement, one or more current `path@sha256#pointer` references, domain and selection scope, manuscript action, migration blocker, and notes.

Status counts are:

- **survives.**
 - **Claims:** 9
- **conditional.**
 - **Claims:** 13
- **changes.**
 - **Claims:** 7
- **rejected.**
 - **Claims:** 12
- **pending.**
 - **Claims:** 6
- **outside_scope.**
 - **Claims:** 6

Manuscript-action counts are 22 `retain_with_rewrite`, 10 `replace`, 9 `remove`, 6 `do_not_add`, and 6 `pending`. These are future migration instructions only. They do not authorize a manuscript edit.

18.1.3 Main classification results

18.1.3.1 Solution, ballot, timing, and beliefs

- The full fixed-pie primitive and proposal-space contract survives: $N \geq 3$, a binary private H type and common prior, unit weak surplus, zero weak outside payoffs, zero direct H agreement benefit, nonnegative named weak payments, residual allocation to the proposer, reabsorption of the conditional H package upon exclusion, and no double receipt by H.
- PBE, the simultaneous sealed ballot, weak-state agenda with `pi_H=0`, the immediate Round-1 opt-out, external H outside options, and formation at equality survive.
- The old pivotal-vote shorthand changes to outcome-signature relevance. The twenty-one-history Gate 0 table changes to the frozen transition, information-set, and relevance registries.
- Terminal payoffs are solved in terminal units, without an internal discount; a terminal continuation is discounted exactly once when imported into a Round-1 incentive comparison.
- The informational principle that uninformed weak ballots do not directly reveal H's type survives conditionally. Hard-coded historical off-path posteriors change to explicit assessment components.
- The full simultaneous-vote incentive comparison and its reduction to a direct package cutoff only under certain implementation already appear together historically and survive together.

18.1.3.2 Equilibrium correspondences and historical reductions

- All-weak approval, zero gifts, and unit proposer retention are rejected as a terminal-majority characterization. Current correspondences admit positive gifts, oversized support, and distinct outcome distributions.
- Historical pooling, low-only, and rejection or continuation motifs survive only as fully proved subclasses. Their exhaustive reduction is rejected.

- Global scalar and uniqueness characterizations for terminal and Round-1 bargaining are rejected. Exact small-domain projections are retained only where the frozen interfaces prove them: the terminal-unanimity attained upper envelope, the active-H terminal-majority unit proposer projection for $3 \leq N \leq 5$ with positive low-type outside option, the exact $N=3$ weak-only continuation, the attainable $N \geq 4$ weak-only upper endpoint, and the $N=3$ Round-1 majority proposer projection under its stated boundary.
- The old boundary Appendix C characterization and its equation/table system do not migrate. Boundary claims must be rebuilt from the current full correspondences.

18.1.3.3 Entry and three distinct nesting notions Entry is assessment-by-assessment on the full $N \geq 3$ common domain. Three different nesting objects are kept separate:

1. pairwise formation-set nesting, exactly determined by the two retained gross values;
2. possible-cost and guaranteed-cost projections within each rule; and
3. universal nesting across every cross-rule assessment pair.

For $N \geq 4$, the possible-cost and guaranteed-cost U-in-M projections survive conditionally, and at each positive cost through the resource ceiling there exists an M-only pair. Universal U-in-M nesting remains pending. The current interfaces also neither prove nor exclude an attained U-only pair, so the old no-U-only statement remains pending rather than rejected.

18.1.3.4 H payoffs and institutional ranking The historical ex-ante outside-option floor survives and is strengthened by the current type-by-type floor. Neither the floor nor common formation supplies an unconditional rule ranking.

Conditional formation-pattern identities survive: neither-form gives H its outside option under both rules; in an exclusive pattern the forming rule weakly benefits H relative to the nonforming rule. When both form, the sign of the retained payoff difference is unidentified without the pair values. The historical weak ordering therefore remains pending.

The separate historical claim that equality in a both-form comparison occurs only in one named majority class is rejected. At $N \geq 4$ and zero entry cost, attained U zero-weak-value and M value-one assessments both form and give H its outside option. This refutes the equality-only clause but does not resolve the weak ordering. Universal H dominance and a universal institutional ranking remain pending.

18.1.3.5 Quarantined and extension material Positive H recognition, endogenous rule choice, delayed-continuation and hybrid exit architectures, old random-proposer formulas, historical OPEC calibration, and closed Goal-3 artifacts are outside the current game or quarantined provenance. They are not current evidence and must not be added to v6 from this matrix. No current No-Cheap-H restriction selects one majority assessment.

18.1.4 Verification and migration block

Run:

```
Rscript --vanilla scripts/verify_pivotal_response_v6_survival_matrix.R
```

Result: **43/43 PASS.**

The verifier checks every source locator, every evidence hash and JSON pointer, the exact status/action locks and counts, all 27 protected hashes, and full topic coverage. Negative mutations confirm that missing evidence, a stronger status, a protected-hash change, an archived formula or branch label in a current replacement, a forbidden historical label assigned to a current PBE, promotion of an old formula, or mutation of either approved comparison dependency blocks the candidate.

The exact nonmigratory candidate was subsequently accepted by two independent read-only reviews and frozen by its review bundle. The protected manuscripts and prior derivation remained byte-identical. No protected-target migration is authorized.

The authoritative matrix has the following final distribution:

Table 10: Claim-level survival classifications after the independent rederivation.

Status	Claims
changes	7
conditional	13
outside_scope	6
pending	6
rejected	12
survives	9

19 Consolidated results and limits

The rederivation changes the substantive object from a small list of scalar candidate outcomes to a full PBE correspondence. Five conclusions organize the result.

1. **Local voting discipline does not generate a global equilibrium selection.** When a vote changes a payoff-relevant outcome with positive probability, yes is prescribed exactly when its expected payoff is weakly higher. When relevance is zero, explicit completions remain part of the assessment. Individual irrelevance is not closed under joint changes, so different full-profile completions can sustain different outcomes.
2. **Terminal correspondences are generally broad.** Under unanimity with an active hegemon, coordinated failure is available from $N \geq 4$; under active-H majority, the recognized-proposer projection is $\{1\}$ for $3 \leq N \leq 5$ when $o_0 > 0$, becomes $[\max\{1 - o_1, 1 - \mu\}, 1]$ when $o_0 = 0$, and is $[0, 1]$ for $N \geq 6$. Weak-only majority is unique only at $N = 3$; from $N \geq 4$ it contains passage, failure, gifts, and identity-distinct allocations. None of these scalar projections replaces the full interface.
3. **Both Round-1 games have a PBE throughout the primitive domain.** The exact characterization is proposal- and history-indexed. Under majority at $N = 3$, with $b = \beta/2$, $A_3 = 1 - b$, $C_3 = (1 - \mu)(1 - o_0) + \mu \cdot b$, and $D_3 = 1 - o_1$, the proposer projection is $\{\max(A_3, C_3, D_3)\}$ if $o_0 > 0$ and $[\max(C_3, D_3), \max(A_3, C_3, D_3)]$ if $o_0 = 0$. A value-one majority PBE exists for every $N \geq 4$. Under unanimity, the familiar pooling, low-only, and continuation motifs survive only as verified subclasses; they do not exhaust the equilibrium correspondence.
4. **Entry is assessment-specific.** For either rule,

$$G_R(\alpha, \mu) = \frac{(1 - \mu)T_W^R(\alpha, 0) + \mu T_W^R(\alpha, 1)}{m},$$

and the weak coalition forms exactly when $G_R \geq \chi$. In both rules, $0 \leq G_R \leq 1/m$. For $N \geq 4$, unanimity attains its lower endpoint zero and majority attains its upper endpoint $1/m$; the other general endpoints are not identified by the frozen interfaces.

5. **The institutional comparison is set-valued.** For a fixed assessment pair, formation costs are $[0, G_R]$, so pairwise U-in-M nesting holds exactly when $G_U \leq G_M$. Possible-cost, guaranteed-cost, and universal cross-assessment nesting are different propositions. For $N \geq 4$, there is an M-only pair at every $0 < \chi \leq 1/m$, and universal M-in-U nesting is false. Universal U-in-M nesting remains pending. The hegemon's payoff is always at least its type-specific outside option, but this floor does not create a universal rule ranking; when both institutions form, the sign is generally indeterminate.

The proof therefore does **not** establish equilibrium uniqueness, a general minimum winning coalition, zero gifts, an all-yes ballot, a global Round-1 cutoff, an exhaustive three-motif reduction, a universal formation nesting, or universal hegemon dominance. Positive hegemon recognition, endogenous rule choice, delayed or hybrid exit, and historical numerical applications are outside this baseline. No result in this workspace has been migrated to the target manuscript.